

Life cycle assessment of strawberry and asparagus production

Client

Peter Gerber, BAFU
Pierryves Padey, BAFU
3003 Bern

Authors

Thomas Kägi, Flora Conte, Tanja Laube, Gavin Roberts, Carbotech AG

Zurich, 14 August 2022

Imprint

Title

Life cycle assessment of strawberry and asparagus production

Commissioned by

Federal Office for the Environment (FOEN), Economics and Innovation Division, CH 3003 Bern. The FOEN is an agency of the Federal Department of the Environment, Transport, Energy and Communications (DETEC)

Contractor

Carbotech AG, Zurich

Author

Thomas Kägi, Flora Conte, Tanja Laube, Gavin Roberts

Project Management / Contact

Thomas Kägi
+41 44 444 20 17
t.kaegi@carbotech.ch

External validation

Andy Eigenmann, myclimate

Note

This study/report was prepared under contract to the Federal Office for the Environment (FOEN). The contractor bears sole responsibility for the content.

Version

1.1

Date

14 August 2022

Table of contents

1	Introduction	4
2	Methodology and procedure	4
2.1	General description of LCA	4
2.2	Procedure of LCA	5
2.3	Objectives and framework conditions	5
2.3.1	Objective	6
2.3.2	Functional unit	6
2.3.3	Application and target group of the study	6
2.3.4	External Review	6
2.3.5	System boundary	6
2.4	Life Cycle Inventories	7
2.4.1	Auxiliary Inventories	8
2.4.2	Cultivation inputs and yields	9
2.4.3	Direct field emissions	13
2.4.4	Land use	17
2.4.5	Summary of life cycle inventory data	18
2.4.6	Further data used for LCIA	18
2.4.7	Data quality	18
2.5	Impact analysis	18
2.6	Assessment of environmental impacts	19
2.7	Uncertainty	20
3	Life Cycle Impact Assessment	20
3.1	Results and discussion for strawberries, delivered to store	20
3.2	Results and discussion for asparagus, delivered to store	22
4	Conclusion	25
5	References	26
A1	Global warming potential	28
A2	EF v3.0 method	29
A3	Table with results	30
A4	Metadata and unit process raw data	33
A4.1	Auxiliary inventories	33
A4.2	Cultivation	41
A5	Review report	55

1 Introduction

There are only a few inventory data for fruit and vegetables in the Federal Office for the Environment (FOEN) database. To raise the awareness of the population about the importance of their choices in the consumption of vegetables and fruit it is necessary to expand the information in this area to include other crops. It was therefore decided to generate the inventory data for strawberries and asparagus.

The project aims to generate new data and update the life cycle inventories (LCI) and the life cycle assessment for the following currently practiced production, transport and packaging scenarios:

Key inventories	Transportation
Strawberries, conventional, heated greenhouse/CH	lorry
Strawberries, conventional, unheated plastic tunnel/CH	lorry
Strawberries, organic, unheated plastic tunnel/CH	lorry
Strawberries, conventional, unheated multi tunnel/ES	lorry
Asparagus, conventional/CH	lorry
Asparagus, conventional/PE	air freight, sea freight
Asparagus, conventional/MX	air freight

2 Methodology and procedure

Today, there is a broad consensus that LCA is the most comprehensive and meaningful method to assess the environmental impacts of products and systems. Therefore, this method is used to determine the environmental impacts of the products mentioned.

2.1 General description of LCA

Life cycle assessment (LCA) is a method for recording and assessing the effects of human activities on the environment and deriving optimisation potential from them. Due to the complexity of nature and the global economic system, it is not sufficient to consider only individual problematic substances or local impacts. The following requirements for the method result from the demand for a comprehensive assessment:

- Consideration of the various environmental impacts as comprehensively as possible
- Consideration of the entire life cycle
- Quantification of the environmental impact
- Evaluation of the different impacts as a basis for decisions
- Scientifically supported to achieve a high level of reliability and acceptance

Life cycle assessment is the method that best meets these requirements today. The results of the life cycle assessment can be used:

- as decision-making support when facing different options
- to account for the relevant impacts
- to determine the main influencing factors
- in strategic planning to identify optimisation potentials
- for the assessment of measures
- as a basis for eco-design
- for the derivation of recommendations for action

2.2 Procedure of LCA

After the problem and the systems to be investigated have been defined, the flows of goods, materials and energy as well as the resource requirements are recorded. Subsequently, the effects on the environment are determined with the help of selected indicators that describe these effects. With the aim of expressing the results with a key figure and thus enabling, or at least facilitating, evaluation, an assessment of the various environmental impacts can be made by weighting them accordingly.

According to ISO 14040/44 (ISO, 2006; ISO/TC, 2006), a life cycle assessment comprises the following steps:

- Defining the objectives and system boundaries (framework conditions)
- Recording the relevant material and energy flows as well as the resource requirements (life cycle inventory)
- Determine the impact on the environment (impact assessment)
- Interpretation of the environmental impacts based on the objectives (interpretation)
- Developing measures (optimisation)

As Figure 1 shows, this is not a linear process, but an interactive process of cognition and optimisation.

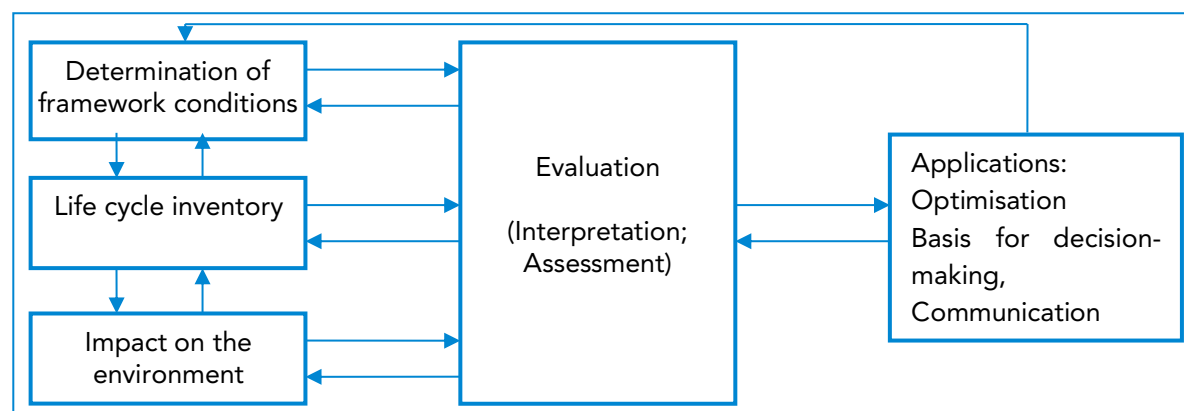


Figure 1: Steps of a life cycle assessment according to ISO 14040/44

The present study follows the ISO 14040 standard; in individual points, such as the use of overall aggregating methods, the present study goes beyond the standard.

2.3 Objectives and framework conditions

The definition of the systems to be examined and compared depends on the objective or question. This results in different framework conditions and system boundaries. The system boundaries define which

processes and upstream processes are taken into account. For example, the time frame of the data used and the environmental impacts to be investigated are defined.

2.3.1 Objective

The life cycle assessment is intended to answer the question of which strawberry or asparagus cultivation is preferable from an ecological point of view. Furthermore, an important goal is to provide life cycle inventories for strawberry and asparagus production. In particular, the following goals should be considered:

- Analysis and comparison of the environmental impact of strawberry cultivation on soil in Switzerland and Andalusia, Spain.
- Analysis and comparison of the environmental impact of asparagus cultivated in Switzerland, Peru and Mexico.

2.3.2 Functional unit

If a product is compared with alternatives, these must provide the same benefit or fulfil the same function. The measure to which the comparison refers to is called the functional unit.

In this study, for the analysis and comparison of the strawberry variants, the following functional unit was defined:

- 1kg of fresh strawberry, delivered to a generic store in Switzerland.
- 1kg of fresh asparagus, delivered to a generic store in Switzerland.

2.3.3 Application and target group of the study

The study is primarily aimed at the client and the interested public.

2.3.4 External Review

An external review of the study was conducted by Andy Eigenmann, myclimate. The review report is integrated in the appendix of the report. The review states that the present study complies with the international standards for life cycle assessments (ISO 14040 and 14044) and the methodological guidelines of the ecoinvent v2 methodological framework (Frischknecht et al., 2007, S. 2). This includes especially:

- Completeness of the documentation. All investigated datasets should be described in the report, and all necessary meta information and flow data should be available for each dataset.
- Consistency with the quality guidelines. It is checked whether the unit processes have been modelled according to the ecoinvent quality guidelines. The quality guidelines cover for example the estimation of transport distances or the calculation of energy demands in the inventory (see chapter 2.4.2).
- Plausibility check of the life cycle inventory data. Selected input and output flows are controlled for plausibility.
- Completeness of inputs and outputs. The completeness of flows is based on the environmental and technical knowledge of the reviewing person. Reviewers are not necessarily technical experts of the processes reviewed. If necessary, they were supported by the person responsible for the report.
- Mathematical correctness of calculations. Selected inputs and outputs are controlled in view of mathematical correctness, e.g. the transport service inputs, the waste heat or CO₂ emissions.

2.3.5 System boundary

This LCA looks at the ecological impacts ‘from cradle to grave’, i.e. from the extraction of the raw materials through the individual steps of the strawberry and asparagus cultivation up to and including disposal of the packaging. In accordance with the life cycle assessment approach, all environmentally relevant processes within the system boundary are recorded and evaluated as far as possible.

The system boundary of the present study thus essentially comprises the following material and energy flows of the subsequent processes and services that are considered relevant (see Figure 2):

- Provision of auxiliary materials and production of strawberries and asparagus
- Transport of raw materials and auxiliary materials, strawberries and asparagus
- Waste along the production chains and from provisioning auxiliary materials and energy
- Provision of energy heat and electricity, energy sources such as oil, natural gas, coal, etc. for the processes involved.

Not considered in this study:

- Packaging of fertilisers and pesticides (not relevant)
- Losses during processing and transportation (not relevant)
- Storage

For all these processes, the impacts due to emissions into soil, air and water as well as the resource requirements (e.g. energy resources or land use) are taken into account.

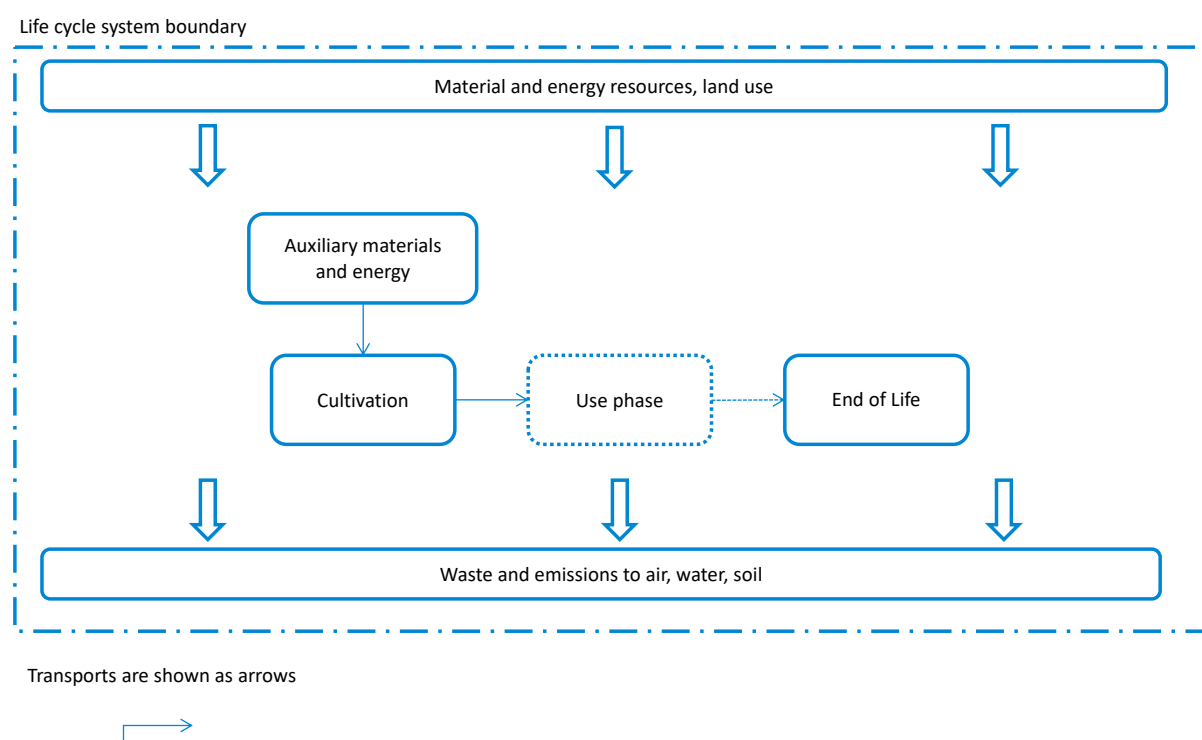


Figure 2: Schematic representation of the processes

Unlike strawberries, asparagus is a perennial crop. Therefore, the data for asparagus cultivation are average values of a multi-year cultivation period: For Switzerland it is the average value of eight years, for Mexico and Peru that of four years.

2.4 Life Cycle Inventories

The report contains information on strawberry and asparagus production. In order to address the complete cultivation system some auxiliary inventories with regard to infrastructure, auxiliary materials and further processing had to be created.

The following inventories were created for the LCI of strawberries and asparagus:

Auxiliary materials and processing

- Strawberry seedling, (organic) for planting, in unheated greenhouse/RER
- Asparagus seedling, for planting, open field/GLO
- Installation for drip irrigation/CH
- Diesel, burned in agricultural machine/CH

Strawberry cultivation

- Strawberries, conventional, heated greenhouse/CH
- Strawberries, conventional, unheated plastic tunnel/CH
- Strawberries, organic, unheated plastic tunnel/CH
- Strawberries, conventional, unheated multi tunnel/ES

Asparagus cultivation

- Asparagus, conventional/CH
- Asparagus, conventional/PE
- Asparagus, conventional/MX

2.4.1 Auxiliary Inventories

2.4.1.1 Strawberry seedling, (organic) for planting, unheated greenhouse/RER

Data for strawberry seedling production were derived from Stoessel et al. (2012) and complemented with approximated fertiliser and pesticide data from tomato seedling production (Kägi et al., 2021). For organic production, no chemical fertiliser and pesticides were used. Direct field emissions were assumed to be similar.

Table 1: Inputs for strawberry seedling production

	Unit	Strawberry seedling production	Strawberry seedling production, organic	Used inventory
N-fertiliser	kg N / p	3.3e-4	0	Average mineral fertiliser, as N, at regional storehouse/RER U
P-fertiliser	kg P2O5 / p	4.2e-4	0	Average mineral fertiliser, as P2O5, at regional storehouse/RER U
K-fertiliser	kg K2O / p	1.6e-3	0	Average mineral fertiliser, as K2O, at regional storehouse/RER U
Organic fertiliser	kg N / p	0	3.3e-4	Manure cattle
Pesticides	kg / p	2.9e-8 1.2e-6	0	Pesticide unspecified Copper oxide
Installation drip irrigation	m3 / p	5.1e-4	5.1e-4	Installation for drip irrigation
Electricity	kWh/p	1.2e-4	1.2e-4	Electricity, low voltage, ENTSO
Peat	kg/p	3e-4	3e-4	Peat, at mine
Plastic tunnel	kg/p	7.7e-7	7.7e-7	Plastic tunnel/ES

2.4.1.2 Asparagus seedling, for planting/RER

Data for asparagus seedling production were taken from [Vazquez-Rowe et al., \(2015\)](#).

Table 2: Material inputs for asparagus seedling production

	Unit	Asparagus, seedling production, 250'000 seedling per ha	Used inventory
Ammonium nitrate	kg N / ha	87	Ammonium nitrate, as N, at regional storehouse
Phosphoric acid	kg / ha	110	Phosphoric acid, fertiliser grade, 70% in H ₂ O, at plant
Potassium sulphate	kg K ₂ O / ha	58	Potassium sulphate, as K ₂ O, at regional storehouse
Magnesium sulphate	kg Mg / ha	2	Magnesium sulphate, at plant
Calcium nitrate	kg N / ha	2	Calcium nitrate, as N, at regional storehouse
Potassium nitrate	kg N / ha	30	Potassium nitrate, as N, at regional storehouse
Pesticides	kg / ha	0.84	Pesticide unspecified
Water	m ³ / ha	3'357	Water, unspecified natural origin
Electricity for irrigation	kWh / ha	1'600	Electricity, low voltage

2.4.1.3 Irrigation, drip

So far, there is no module for drip irrigation in the UVEK database. Therefore, we created a drip irrigation inventory for Swiss conditions. Amounts and materials for this inventory were derived from the Handbook on pressurized irrigation techniques (Phocaides, 2007).

2.4.1.4 Diesel, burned in agricultural machine

This inventory is based on transport, tractor and trailer from the UVEK database but the functional unit was converted from tkm to the corresponding demand of diesel in kg.

2.4.2 Cultivation inputs and yields

2.4.2.1 Yields

Yields for strawberry cultivation in Switzerland were provided by the Swiss Fruit Association ([Schweizer Obstverband, 13. April 2022](#)). Yield for strawberry cultivation in Spain was taken from the LCA on environmental footprint of cultivating strawberry in Spain (Romero-Gómez & Suárez-Rey, 2020).

Yield for asparagus cultivation in Switzerland was modelled according to the factsheet for asparagus cultivation of Strickhof (2016) is an average value for a cultivation period of eight years (year 1 and 2: 0 t / ha, year 3 and 8: 4.5 t / ha, year 4 and 7: 4.75 t / ha, year 5 and 6: 5 t / ha). Yield for asparagus from Peru was taken from [Vazquez-Rowe et al., \(2015\)](#) and is an average value for a cultivation period of four years. For Mexico, the yield is based on a newspaper report and represents the average yield of different regions (Godina Gonzalez, 2018).

Table 3: Yields for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated tunnel/CH	Strawberries, organic, unheated tunnel/CH	Strawberries, conventional, unheated tunnel/ES	multi
Source		(Schweizer Obstverband, 2022)	(Schweizer Obstverband, 2022)	(Schweizer Obstverband, 2022)	(Romero-Gómez & Suárez-Rey, 2020)	
Yield	t / (ha*a)	52	14	12	55	

Table 4: Yields for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX
Source		(Strickhof, 2016)	(Vazquez-Rowe et al., 2015)	(Godina Gonzalez, 2018)
Yield	t / (ha*a)	3.6	6.9	6.2

2.4.2.2 Material inputs

Strawberries

The material input on strawberry cultivation includes data on materials needed to install the tunnels and auxiliary systems, fertilisers, pesticides and water for irrigation.

As for Switzerland, data for auxiliary materials for strawberry cultivation in plastic tunnel were taken from [Valiante et al. \(2019\)](#), for cultivation in greenhouse conditions they were adopted from the inventory of tomatoes in heated greenhouse. The amount of water input was modelled within the project ‘World Food LCA database (WFLDB)’ and is representative for strawberry production in Switzerland under heated greenhouse conditions and non-heated greenhouse conditions, respectively. Data for fertiliser inputs in conventional strawberry cultivation refer to the standards for fertilisation of Agroscope (Carlen & Ançay, 2017). Pesticide inputs were derived from the Greenpeace study on pesticides in Swiss strawberries (Schenkel et al., 2016). Since no data on fertiliser usage was available for organic strawberry production in Switzerland, inputs were taken from various interviews with strawberry producers ([anonymous, April 2022](#)). Natural products for the treatment of pests and diseases in organic farming are not considered in this study.

For strawberry cultivation in Spain input data for fertiliser, water and auxiliary materials (pipes, pumps, mulch) were taken from [Romero-Gómez & Suárez-Rey \(2020\)](#). Pesticide inputs were calculated on the basis of the laboratory findings of the Chemical and Veterinary Investigation Office Stuttgart (Dr. Benkenstein, 2021).

Asparagus

Data for asparagus cultivation in Switzerland were derived from [Soode et al. \(2015\)](#) and are average values of a cultivation period of eight years. This does not include the data for fertilising, these are taken from Neuweiler and Krauss ([2017](#)) and Feller ([2019](#)).

All data for Peru were derived from Vazquez-Rowe et al. (2015) and are average values of a cultivation period of four years. For Mexico, the same data was used and the country-specific data was adjusted.

The data on pesticides for asparagus cultivation in Switzerland and Mexico were calculated on the basis of the laboratory findings of the Chemical and Veterinary Investigation Office Stuttgart (Dr. Benkenstein, 2021). For all countries, source for the amount of seedlings is the Swiss Vegetable Producers Association (2022).

Table 5: Material inputs for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/ CH	Strawberries, conventional, unheated plastic tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated multi tunnel/ES	Used inventory
N-fertiliser	kg N / ha	220	100	0	275	Average mineral fertiliser, as N, at regional storehouse/RER U
P-fertiliser	kg P ₂ O ₅ / ha	104	34	0	247	Average mineral fertiliser, as P ₂ O ₅ , at regional storehouse/RER U
K-fertiliser	kg K ₂ O / ha	247	121	0	375	Average mineral fertiliser, as K ₂ O, at regional storehouse/RER U
Magnesia	kg / ha	50	20	0	0	Magnesia, at plant
Organic fertiliser	kg N / ha	0	0	150	0	Manure cattle
Pesticides	kg / ha	1.59	1.59	0	5.95	Pesticide unspecified
Water, tap water	m ³ / ha	822	493	493	6'447	Tap water, unspecified natural origin
Water, desalinated water	m ³ / ha	0	0	0	1'321	Tap water, desalinated sea water/ES U
Plastic equipment	kg / ha	100	508	508	526	Polypropylene/Polyethylene, granulate + extrusion
Steel equipment	kg / ha	0.1	0	0	1.5	Steel, low alloyed + section bar rolling
Strawberry seedlings	p	45'000	45'000	45'000	45'000	Strawberry seedling, unheated greenhouse, at production site

Table 6: Material inputs for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX	Used inventory
N-fertiliser	kg N / ha	108	0	0	Average N fertiliser
P-fertiliser	kg P ₂ O ₅ / ha	36	0	0	Average P fertiliser
K-fertiliser	kg K ₂ O / ha	126	0	0	Average K fertiliser
Ammonium nitrate	kg N / ha	0	63.8	81.5	Ammonium nitrate, as N, at regional storehouse
Phosphoric acid	kg / ha	0	242.3	241.5	Phosphoric acid, fertiliser grade, 70% in H ₂ O, at plant
Potassium sulphate	kg K ₂ O / ha	0	149.7	117.0	Potassium sulphate, as K ₂ O, at regional storehouse
Magnesium sulphate	kg Mg / ha	26	27.5	27.6	Magnesium sulphate, at plant

Calcium nitrate	kg N / ha	0	8.3	9.2	Calcium nitrate, as N, at regional storehouse
Boric acid	kg / ha	0	5.9	8.5	Boric acid, anhydrous, powder, at plant
Urea ammonium nitrate	kg N / ha	0	90.0	62.8	Urea ammonium nitrate, as N, at regional storehouse
Potassium nitrate	kg N / ha	0	22.2	36.7	Potassium nitrate, as N, at regional storehouse
Organic fertiliser	kg / ha	0	39.5	68.0	Poultry manure as proxy for guano (PE/MX)
			49	34.1	Vinasse as proxy for liquid fertiliser
Pesticides	kg / ha	5.8	9.9	6.1	Pesticide unspecified
Water	m ³ / ha	19	10'107	10'107	Water, unspecified natural origin
Plastic equipment	kg / ha	53	0	0	Polyethylene granulate + extrusion (foil)
Asparagus seedlings	P / (ha * a)	2875	5'750	5'750	Asparagus seedling, open field, at production site

2.4.2.3 Energy and machinery use

Strawberries

Heat demand for heated greenhouse in Switzerland was reported by [Energie-Agentur der Wirtschaft \(2022\)](#) and corresponds to the average value of two Swiss plants from the years 2017 to 2020. Typically, greenhouses are heated from March to May. Electricity use for heating system was estimated as 1 % of heat demand based on data for heating systems, further electricity use was estimated based on the inventory of tomatoes in heated greenhouse.

Electricity use for all processes in multi tunnel cultivation in Spain and machinery use were derived from [Romero-Gómez & Suárez-Rey \(2020\)](#) and applied for Switzerland. Energy and diesel consumption for multi tunnel cultivation in Spain was taken from [Romero-Gómez & Suárez-Rey \(2020\)](#), for Switzerland from [Valiante et al. \(2019\)](#).

Asparagus

Energy data for asparagus cultivation were derived from [Soode et al. \(2015\)](#) for Switzerland, from [Vazquez-Rowe et al. \(2015\)](#) for Peru and for Mexico. Except for machinery use, these data refer to [Romero-Gómez & Suárez-Rey \(2020\)](#).

Table 7: Energy and machinery use for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated plastic tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated multi tunnel/ES	Used inventory
Heat for heating	kWh / ha	320'000	0	0	0	Heat, natural gas or heat from waste
Electricity, for heating unit	kWh / ha	3'200	0	0	0	Electricity, low voltage
Electricity for irrigation and machinery	kWh / ha	189	114	114	1'789	Electricity, low voltage
Diesel	kg / ha	0	44.5	44.5	42.3	Diesel, burned in agricultural machine

Table 8: Energy and machinery use for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX	Used inventory
Electricity for irrigation and machinery	kWh / ha	869	3'665	3'417	Electricity, low voltage
Diesel	kg / ha	445	66.3	63.9	Diesel, burned in agricultural machine

2.4.3 Direct field emissions

Strawberries

For Spain, the fertiliser emissions were taken from [Romero-Gómez & Suárez-Rey \(2020\)](#). All other direct field emissions were calculated according to the methodology described in Nemecek and Schnetzer (2011). Nitrate leaching was calculated according to the SQCB-model.

Asparagus

All direct field emissions were calculated according to the methodology described in Nemecek and Schnetzer (2011). Nitrate leaching was calculated according to the SQCB-model. The quantities are valid for the whole period (total).

2.4.3.1 Emissions to air

Table 9: Emissions to air for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated tunnel/ES
NH ₃ in air	kg NH ₃ / ha*a	1.15E+01	5.23E+00	43.7E+01	1.02E+01
N ₂ O in air	kg N ₂ O / ha*a	3.80E+00	1.96E+00	1.27E+00	2.75E+00
NO _x in air	kg NO _x / ha*a	8.00E-01	4.10E-01	2.66E-01	7.15E+00
Water to air	kg H ₂ O / ha*a	7.42E+05	4.45E+05	4.45E+05	6.56E+06

Table 10: Emissions to air for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX
CO ₂ in air	kg CO ₂ / ha*a	0	6.06E+01	4.23 E+01
NH ₃ in air	kg NH ₃ / ha*a	5.65E+00	1.23E+01	1.11E+01
N ₂ O in air	kg N ₂ O / ha*a	2.00E+00	3.31E+00	3.49E+00
NO _x in air	kg NO _x / ha*a	4.20E-01	7.00E-01	7.30E-01
Water to air	m ³ / ha*a	1.13E+01	4.69E+03	4.68E+03

2.4.3.2 Emissions to soil

Table 11: Emissions to soil for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated tunnel/ES
Cd	kg / ha	0	0	0	1.90E-02
Cu	kg / ha	3.36E-03	3.80E-05	0	0
Azoxystrobin	kg / ha	2.00E-02	2.00E-02	0	6.00E-02
Benzamide	kg / ha	0	0	0	1.10E-01
Bifenazat	kg / ha	0	0	0	2.00E-02
Bupirimate	kg / ha	4.00E-02	4.00E-02	0	0
Chlorate	kg / ha	0	0	0	5.25E+00
Chlorpyrifos- methyl	kg / ha	0	0	0	1.00E-02

Cyprodinil	kg / ha	2.30E-01	2.30E-01	0	7.00E-02
Difenoconazole	kg / ha	4.00E-02	4.00E-02	0	0
Dimethomorph	kg / ha	0	0	0	4.00E-02
Etoxazole	kg / ha	1.00E-02	1.00E-02	0	1.00E-02
Fenhexamid	kg / ha	5.30E-01	5.30E-01	0	0
Fludioxonil	kg / ha	1.00E-01	1.00E-01	0	5.00E-02
Fluopyram	kg / ha	8.00E-02	8.00E-02	0	1.20E-01
Fosetyl	kg / ha	0	0	0	1.00E-01
Indoxacarb	kg / ha	0	0	0	1.00E-02
Isopropyl hyradzincarboxylat e	kg / ha	0	0	0	2.00E-02
Kresoxim-methyl	kg / ha	9.00E-02	9.00E-02	0	0
Myclobutanil	kg / ha	2.00E-02	2.00E-02	0	2.00E-02
Penconazole	kg / ha	0	0	0	1.00E-02
Pendimethalin	kg / ha	0	0	0	1.00E-02
Phenmedipham	kg / ha	1.20E-01	1.20E-01	0	0
Quinoxifen	kg / ha	1.00E-02	1.00E-02	0	0
Sodium chlorate	kg / ha			0	5.25E+00
Spinosad	kg / ha	0	0	0	2.00E-02
Thiacloprid	kg / ha	1.30E-01	1.30E-01	0	0
Trifloxystrobin	kg / ha	1.60E-01	1.60E-01	0	1.60E-01

Table 12: Emissions to soil for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX
Cd to soil	kg / ha	2.00E-03	0	1.90E-03
Cu to soil	kg / ha	0	0	0
Zn to soil	kg / ha	0	0	0
Pb to soil	kg / ha	0	0	0
Ni to soil	kg / ha	7.40E-04	3.10E-03	4.70E-03
Cr to soil	kg / ha	0	0	0

Hg to soil	kg / ha	0	6.60E-06	8.30E-06
Azoxystrobin	kg / ha	1.00E-02	0	0
Boscalid	kg / ha	4.00E-02	0	5.00E-02
Chlorate	kg / ha	5.58E+00	0	5.17E+00
Chlorpyrifos	kg / ha	0	8.20E+00	0
Cyprodinil	kg / ha	2.00E-02	0	0
Dimethomorph	kg / ha	1.25E-03		
Flonicamid	kg / ha	0	0	1.00E-02
Fludioxonil	kg / ha	1.00E-02	0	2.50E-02
Fosetyl	kg / ha	6.00E-02	0	0
Imidacloprid, Olefin	kg / ha	0	0	2.50E-02
Metalaxyl	kg / ha	0	0	2.50E-02
Methomyl	kg / ha	0	1.50E+00	0
Metribuzin	kg / ha	5.20E-01	0	0
Myclobutanil	kg / ha	1.25E-03		
Sulfur	kg / ha	0	5.00E-02	0
Pendimethalin	kg / ha	5.00E-03		
Quinoxifen	kg / ha	1.25E-03		
Thiacloprid	kg / ha	2.50E-03		
Trifloxystrobin	kg / ha	3.75E-03		
Tebuconazole	kg / ha	0	1.75E-01	0
Glyphosate	kg / ha	0	0	8.30E-01

2.4.3.3 Emissions to water

Table 13: Emissions to water for strawberry cultivation

	Unit	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated tunnel/ES	multi
Nitrate in ground water	kg NO ₃ / ha	1.29E+02	7.50E+01	2.10E+02	8.09E+01	
Phosphorus in ground water	kg P / ha	7.00E-02	7.00E-02	7.00E-02	7.00E-02	

Phosphorus in surface water	kg P / ha	7.80E-01	7.50E-01	8.20E-01	8.40E-01
Cd to water	kg / ha	4.5E-05	3.8E-05	2.9E-05	4.8E-05
Cu to water	kg / ha	3.2E-03	2.9E-03	3.6E-03	3.4E-03
Zn to water	kg / ha	1.6E-02	9.4E-03	2.9E-02	2.1E-02
Pb to water	kg / ha	1.2E-04	6.1E-05	3.1E-04	1.9E-04
Ni to water	kg / ha	7.8E-06	4.7E-06	1.2E-05	9.7E-06
Cr to water	kg / ha	1.9E-02	1.7E-02	1.8E-02	2.0E-02
Hg to water	kg / ha	0	0.0E+00	1.3E-06	0
Water to water	m ³ / ha	3.32E+01	3.54E+01	3.72E+01	1.16E+03

Table 14: Emissions to water for asparagus cultivation

	Unit	Asparagus, conventional/CH	Asparagus, conventional/PE	Asparagus, conventional/MX
Nitrate in ground water	kg NO ₃ / ha	1.15E+02	1.63E+02	1.75E+02
Phosphorus in ground water	kg P / ha	7.00E-02	7.00E-02	7.00E-02
Phosphorus in surface water	kg P / ha	7.50E-01	7.40E-01	7.50E-01
Water to water	m ³ / ha	4.39E+00	5.41E+03	1.01E+04
Cd to water	kg / ha	3.8E-05	3.9E-06	3.8E-05
Cu to water	kg / ha	2.9E-03	2.7E-03	2.7E-03
Zn to water	kg / ha	9.8E-03	1.2E-02	1.4E-02
Pb to water	kg / ha	6.4E-05	9.3E-05	9.4E-05
Ni to water	kg / ha	4.9E-06	8.3E-06	8.8E-06
Cr to water	kg / ha	1.7E-02	1.0E-02	1.6E-02
Hg to water	kg / ha	0	1.6E-07	1.9E-07

2.4.4 Land use

Land use is already included in the greenhouse and tunnel infrastructure with regard to the strawberry cultivation. It corresponds to a little more than 10'000 m² per ha and year.

With regard to asparagus, duration of one cropping cycle of asparagus is 12 months therefore 10'000 m² per ha and year is used.

2.4.5 Summary of life cycle inventory data

The life cycle inventories for the newly modelled and updated processes are provided as unit process raw data in the EcoSpold v1 format (unit process in SimaPro). The electronic data includes the full EcoSpold v1 documentation.

For each investigated process, two types of tables (X-Process and X-Exchange) are provided. For better readability of the whole report, these data are listed in the Annex.

2.4.6 Further data used for LCIA

Apart from the newly created life cycle inventories, the following data were used and connected to the respective background data:

Transportation

- 150 km by lorry 32-40t for Swiss strawberries and Swiss asparagus from farm to regional centre in Switzerland and 100 km by lorry 16-32t for distribution to store.
- 2'000 km by lorry 32-40t for strawberries from Andalusia to regional centre in Switzerland and 100 km by lorry 16-32t for distribution to store.
- 260 km by lorry 32-40t for asparagus from Peru from plant to port or airport in Callao, 10'673 km air freight to Switzerland (option: 15'277 km sea freight to Rotterdam and 700 km by lorry 32-40t to regional centre in Switzerland) and 150 km by lorry 16-32t for transport to storage and distribution to store.
- 260 km by lorry 32-40t for asparagus from Mexico from plant to airport, 9'516 km air freight to Switzerland, and 150 km by lorry 16-32t for transport to storage and distribution to store.

Packaging

We assumed the following packaging materials:

- Strawberries: boxes on 91.85 g cardboard and 24.6 g PE foil / kg of product
- Swiss asparagus: boxes on 114.8 g cardboard and 25.667 g PE foil / kg of product
- Imported asparagus: boxes on 225 g polypropylene and 29.7 g PE foil / 5 kg of product

2.4.7 Data quality

The data quality of the relevant data is generally good. Yields, material inputs, field work and direct field emissions were updated for this study. Other inputs and outputs which have not been updated during this study are usually of very low relevance for the calculated environmental impacts.

2.5 Impact analysis

In this step, the life cycle inventory is assessed with regard to the impact on the environment. The calculation of the impact analysis includes the following two sub-steps:

- Classification (classification of substances from the life cycle inventory with regard to their effects)
- Characterisation (calculation of the effects on the environment):

In this process, the individual substances are weighted against each other according to their damage potential with respect to a lead substance. This results in the damage potentials with regard to a specific environmental impact. In the case of the global warming potential, CO₂ is used as the lead substance and contributions from other greenhouse gases such as methane and nitrous oxide are converted into CO₂ equivalents.

On midpoint level (impact categories) only the Global Warming Potential is presented in this study (see Annex).

2.6 Assessment of environmental impacts

The result of the impact assessment is a compilation of various indicators, each of which describes one aspect of the environmental impact. In order to obtain a well-founded basis for decision-making, the various impacts can be weighted and summarised in a key figure. The weighting of different environmental impacts is a process in which values are incorporated and which is therefore supported as widely as possible to ensure a high level of acceptance.

In the context of this study, the aggregated ecopoints of the ecological scarcity method 2021 (Frischknecht et al., 2021) was used. The results are expressed in eco-points ('Umweltbelastungspunkte' – UBP). The assessment using the ecological scarcity method was developed with the cooperation of the Federal Office for the Environment and is well established in Switzerland. This method was chosen because it takes into account the environmental situation as well as the environmental goals of Switzerland for the assessment (see Figure 3) and is thus broadly supported in terms of values. With regard to the use of the overall aggregating methods, the present study deviates from ISO standard 14040. Since the assessment of the various environmental impacts depends on value standards, these overall aggregating methods are partly rejected. It should be noted that also the selection of specific environmental impacts is subjective. If only a part of the impacts is considered, e.g. only the carbon footprint, this is virtually the same as weighting the other impacts with zero. The consideration of the individual impact categories can certainly be helpful, e.g. to determine the causes of specific impacts and to work out possible optimisation potentials. However, individual environmental aspects must not be excluded as a basis for decision-making or for considering the overall environmental impacts. For this, overall aggregating assessment methods are not only helpful but also necessary (Kägi et al., 2016) in order to ensure the validity of the results. As a second approach, the EF v3.0 method (European Commission. Joint Research Centre., 2019) was used. The EF v3.0 provides 16 different impact categories and a normalisation and weighting scheme to derive the total environmental footprint (single score).

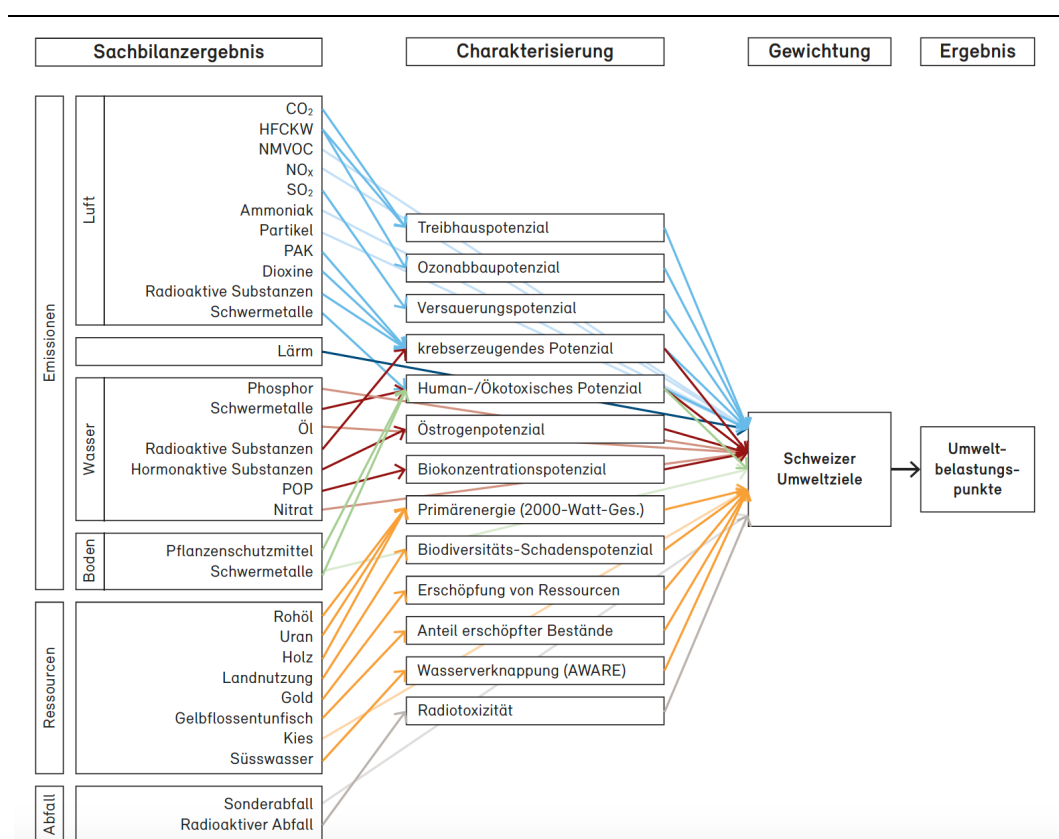


Figure 3: Basic scheme of the ecological scarcity method (graphic from Frischknecht u. a., 2021)

2.7 Uncertainty

For each of the used data inputs in all the inventories, data uncertainty was considered by using the pedigree matrix provided in Simapro. The pedigree matrix translates qualitative aspects with regard to completeness, temporal correlation, geographic correlation as well as further technological correlation into a quantitative uncertainty value. Based on this data uncertainty, a monte carlo analysis is performed in order to address the overall uncertainty of the results.

3 Life Cycle Impact Assessment

Only the environmental footprint according to the ecological scarcity method 2021 is shown and discussed. The carbon footprint (see annex 1)) shows similar results, the environmental footprint according to the EF v3.0 method (see annex 2) shows some differences.

3.1 Results and discussion for strawberries, delivered to store

Figure 4 shows the environmental footprint of the different strawberry variants.

Comparison of variants

Heated greenhouse strawberries from Switzerland have by far the highest environmental impact due to the fossil energy demand for heating, even though the yield is much higher than in unheated plastic tunnels (52 t / ha compared to 14 t / ha in conventional and 12 t / ha in organic cultivation).

The results for the strawberries cultivated in unheated plastic tunnel are in a similar range. Conventional strawberries in unheated plastic tunnels in Andalusia, Spain have a much higher yield per ha (55 t / ha compared to 14 t / ha and 12t / ha) than unheated conventional or organic Strawberries in Switzerland and show lower impact contribution with regard to fertilisers, pesticides or other cultivation processes. However, the impact from transportation is much higher for the Spanish strawberries. The greenhouse infrastructure contributes similarly even though the yield is much higher in Spain because the Spanish unheated greenhouse infrastructure is much more intensive than the Swiss ones.

Process contribution

Considering heated greenhouse cultivation based on natural gas, the highest shares come from the combustion of natural gas and the infrastructure of the greenhouse itself. All other processes are of low relevance.

For strawberries cultivated in unheated plastic tunnel, infrastructure is the main contributor to the environmental impact. With the Swiss variants, infrastructure is followed by fertilizing, further cultivation processes and packaging who show equal shares (about 20%). All other processes – such as pesticides, water scarcity and transportation – play a minor role. In the case of strawberries from Spain, after infrastructure transport and packaging cause the highest impact and are of more relevance than the remaining processes.

Considering the cultivation in Andalusia, Spain, it becomes obvious that water scarcity is not as relevant anymore even for water-intensive crops in regions with high water scarcity compared to former LCA studies that used the ecological scarcity 2013 method. In this respect, the new method is very different from the 2013 method.

Pesticides contribute little to the environmental footprint in this evaluation, although a great deal of pesticides are used in conventional strawberry cultivation.

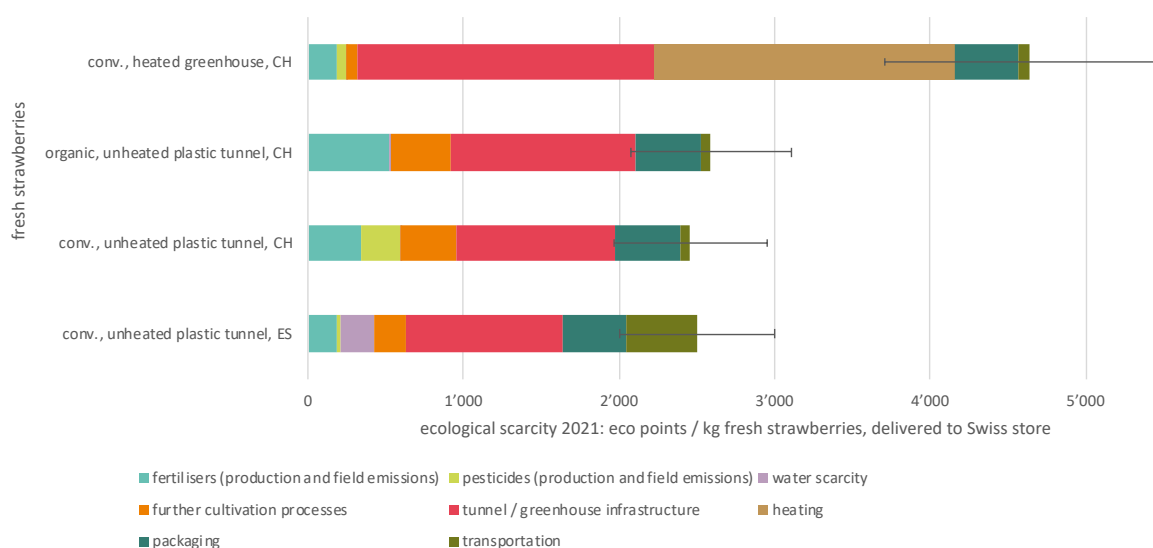


Figure 4: Process contributions to the environmental impact for strawberries, distributed to generic Swiss store

Relevant impact categories

With all variants, global warming is the most relevant impact category. Especially for the heated systems due to CO₂-emissions from natural gas and the unheated system in Spain due to higher demand of auxiliary materials and the desalination of sea water.

Other air emissions and water pollutants also show relevance. So does land use with the Swiss variants due to lower yields.

The water resources are of some relevance in Spain, where water scarcity is high. It should be noted that the average water scarcity factor for Spain was used, as there is only one water flow for Spain. The fact is that most strawberries come from Almeria, where water scarcity is even much higher. However, an inspection of the ecological scarcity methodology shows that the maximum water scarcity factor would be 25 % higher than that for Spain. I.e. even if the maximum water scarcity was calculated in the ecological scarcity method, the conclusions would not be different.

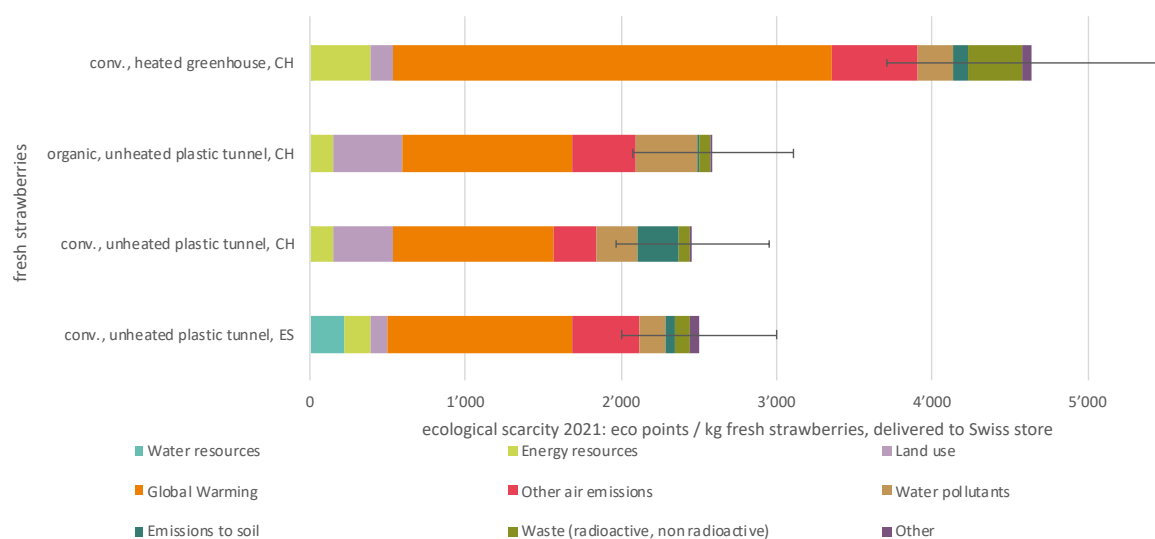


Figure 5: Environmental impact for strawberries, distributed to generic Swiss store

3.2 Results and discussion for asparagus, delivered to store

Figure 6 shows the environmental footprint of the different asparagus variants.

Comparison of variants

The lowest environmental footprint is shown by asparagus from Swiss production. It is significantly lower than with all other variants. This is mainly due to the short transport distance and the mode of transport. Transport makes the big difference between asparagus from Switzerland and asparagus from Mexico.

Asparagus from Peru show a worse outcome than asparagus from Mexico because of the pesticides. The use of Chlorpyrifos makes the difference. If it weren't for the Chlorpyrifos, the result of asparagus from Peru by sea would be similar to Mexican asparagus.

The highest environmental footprint is shown by asparagus from Peru which are imported by air freight.

Process contribution

With asparagus from Peru, pesticides contribute most to the environmental footprint due to the very high environmental toxicity and the risk to human health of Chlorpyrifos. If only this one pesticide were replaced, the outcome would be different. The fact that precisely this pesticide is frequently used in Peru has been confirmed not only by the data from studies but also by laboratory findings.

Considering transport from overseas, once again the high environmental impact of flights is confirmed. With asparagus from Mexico, air freight is responsible for approximately 80% of the impacts, with asparagus from Peru it's because of the high share of pesticides 'only' a little over 40%. All other processes play a secondary role.

With regard to asparagus from Switzerland, the footprint is dominated by the further cultivation processes which cause half of the total impacts, followed by the impacts of fertilization.

Packaging is with all variants comparatively less relevant, but is slightly more influential for asparagus from Switzerland than for asparagus from overseas.

Water scarcity only shows a certain relevance with asparagus from Mexico. It should be noted that the average water scarcity factor for Peru and Mexico was used, as there is only one water flow each for Peru and Mexico. The fact is that especially in Peru, asparagus is mainly grown in arid areas, where water scarcity is much higher. An inspection of the ecological scarcity methodology shows that the maximum water scarcity factor would be 4 times higher than that for Peru. I.e. even if the maximum water scarcity was calculated in the ecological scarcity method, the conclusions would not be different. However, water scarcity would then be similarly relevant as the fertilisers.

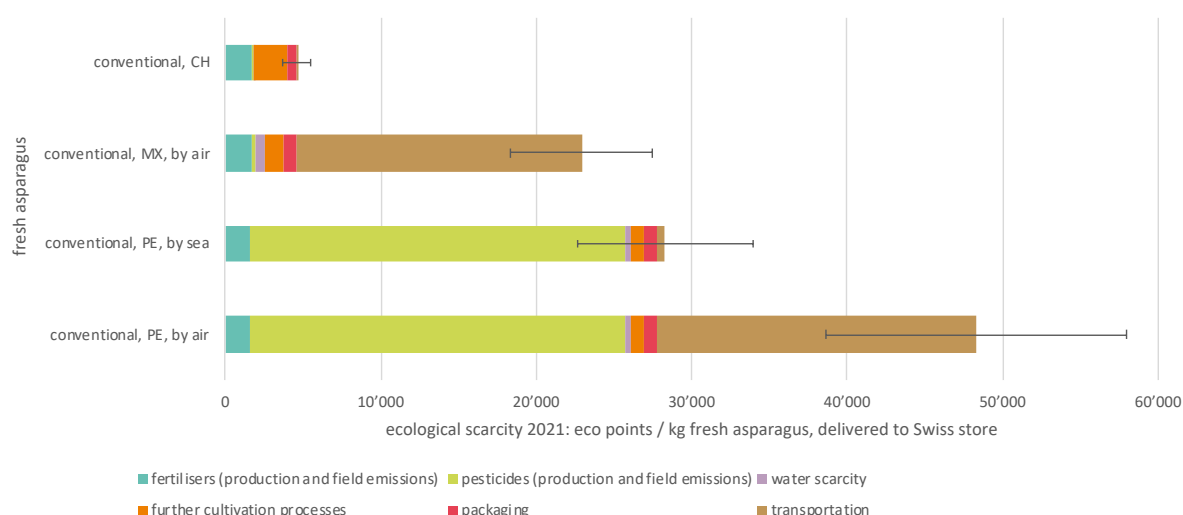


Figure 6: Process contributions to the environmental impact for asparagus, distributed to generic Swiss store

Relevant impact categories

Considering asparagus from Peru, the most relevant impact category is emissions to soil (pesticides). As for the variant imported by air freight, global warming is a major impact category too.

Regarding asparagus produced in Switzerland, global warming (mainly stemming from diesel burnt in agricultural machinery) is the most dominant impact category followed by water pollutants (nitrate leaching) and other air emissions.

Due to the emissions from air freight, global warming is the undisputed leading impact category with asparagus from Mexico. All other impact categories show very little relevance.

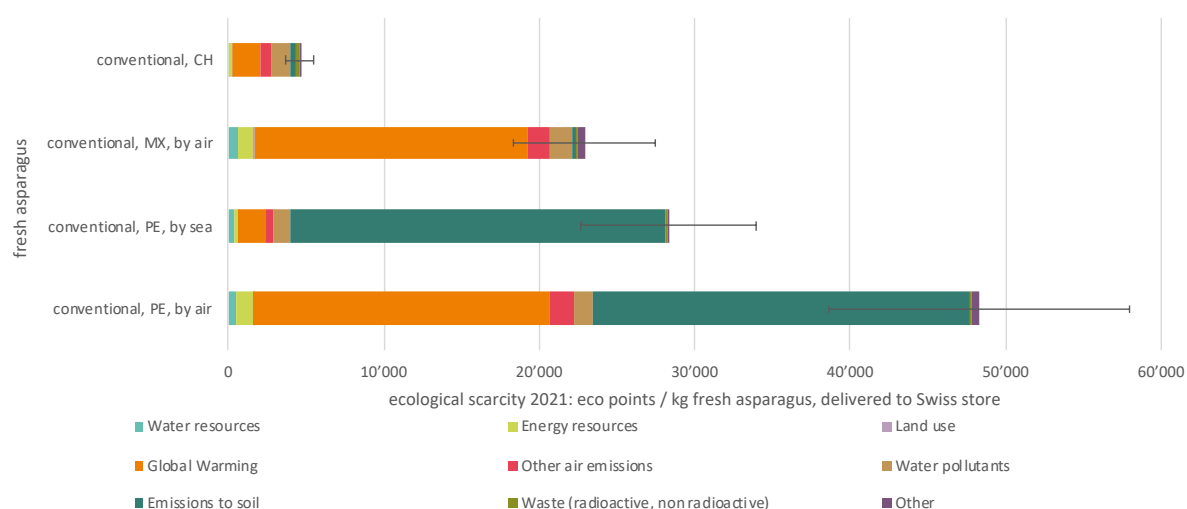


Figure 7: Environmental impact for asparagus, distributed to generic Swiss store

Evaluations according to other methods

For strawberries, the carbon footprint according to IPCC 2021 (see annex 1) correlates with the ecological footprint based on the ecological scarcity method. For asparagus, Peruvian asparagus transported by ship has a similar carbon footprint as Swiss asparagus, as pesticide emissions are not taken into account.

The environmental footprint according to the EF v3.0 methods (see annex 2) does not correlate well with the environmental footprint based on the ecological scarcity method. For strawberry production the difference between heated and unheated strawberry production is less clear due to higher importance of water scarcity (ES production) and acidification and eutrophication (organic CH) in the EF method compared to the ecological scarcity method. For asparagus production, the Peruvian asparagus transported by ship shows a lower impact than Mexican asparagus imported by air because of lower importance of the pesticide emissions.

4 Conclusion

Due to the high efficiency, strawberries from Spain have a similar environmental impact as those from unheated production in Switzerland. Fruits cultivated in heated greenhouse using fossil heating energy have by far the highest environmental impact. For strawberries from tunnel production, cultivation is more impact-relevant than transport and packaging.

Regarding asparagus, the LCA confirms the results of other studies: fresh, seasonal and domestic vegetables are the best choice. In the case of asparagus from overseas, from an environmental point of view those imported by ship are to be preferred to those imported by airplane. With import by air freight, impacts of transportation dominate the ecological footprint.

In the case of pesticides, it is evident that the type of pesticide used plays a major role. Many substances that are banned in Europe are applied in other countries as the regulations are less strict. But even within Europe banned active substances are detected time and again. The use of pesticides is non-transparent with all conventional cultivations. Consumers who want to be on the safe side, choose organic options.

So from an environmental point of view, local production should be promoted for several reasons, e.g. the reduction of emissions due to short transportation routes and the better regulation of pesticides use, especially compared to overseas countries. The use of renewable energy should also play a role in future cultivation.

With regard to the used impact method ecological scarcity 2021 the most obvious difference to the former method is the very low relevance of water scarcity.

5 References

anonymous. (2022, April). *Fertilising of organic strawberries in Switzerland, personal interviews with producers* [Persönliche Kommunikation].

Carlen, C., & Ançay, A. (2017). 14/ Düngung von Beerenkulturen. *Agroscope*, 12.

Dr. Benkenstein, A. (2021). *Pflanzenschutzmittelrückstände in Spargel und Erdbeeren, Untersuchungen Chemisches und Veterinäruntersuchungsamt Stuttgart, 2015 bis Juni 2021*. https://www.ua-bw.de/pub/beitrag.asp?subid=0&Thema_ID=5&ID=3439&Pdf=No&lang=DE

Energie-Agentur der Wirtschaft. (2022, Februar 10). *Energieverbrauch von Erdbeeren im Gewächshaus CH* [Persönliche Kommunikation].

European Commission. Joint Research Centre. (2019). *Suggestions for the update of the Environmental Footprint Life Cycle Impact Assessment: Impacts due to resource use, water use, land use, and particulate matter*. Publications Office. <https://data.europa.eu/doi/10.2760/78072>

Feller, C. (2019). *Düngung im Spargelanbau, Beschreibung von Nährstoffmangelsymptomen*. Leibniz-Institut für Gemüse- und Zierpflanzenbau.

Frischknecht, R., & Büsser Knöpfel, S. (2013). *Ökofaktoren Schweiz 2013 gemäss der Methode der Ökologischen Knappheit—Methodische Grundlagen und Anwendung auf die Schweiz* (Nr. 1330; Umwelt-Wissen, S. 256). Bundesamt für Umwelt.

Frischknecht, R., Dinkel, F., Braunschweig, A., Ahmadi, M., Kägi, T., Krebs, L., Stettler, C., Zschokke, M., Itten, R., & Stucki, M. (2021). *Ökofaktoren Schweiz 2021 gemäss der Methode der Ökologischen Knappheit—Methodische Grundlagen und Anwendung auf die Schweiz* (S. 260). Bundesamt für Umwelt.

Frischknecht, R., Jungbluth, N., Althaus, H. J., Doka, G., Dones, R., Heck, T., Hirschier, R., Nemecek, T., Rebitzer, G., & Spielmann, M. (2007). *Overview and Methodology. Ecoinvent report No. 1, v2.0*.

Godina Gonzalez, A. (2018). *Mexico has tripled the area devoted to asparagus in the last decade*. <https://www.freshplaza.com/article/2187819/mexico-has-tripled-the-area-devoted-to-asparagus-in-the-last-decade/>

ISO. (2006). *ISO 14040:2006 Environmental management—Life cycle assessment—Principles and framework*.

ISO/TC. (2006). *Environmental management—Life cycle assessment—Principles and framework*. International Organization for Standardization.

Kägi, T., Dinkel, F., Frischknecht, R., Humbert, S., Lindberg, J., De Mester, S., Ponsioen, T., Sala, S., & Schenker, U. W. (2016). Session “Midpoint, endpoint or single score for decision-making?”—SETAC Europe 25th Annual Meeting, May 5th, 2015. Conference Session Report. *Int J Life Cycle Assess*, 21(1), 129–132. <https://doi.org/10.1007/s11367-015-0998-0>

Kägi, T., Glimmann, M., Conte, F., & Roberts, G. (2021). *Life cycle assessment of tomatoes and green beans production*. on behalf of FOEN.

Nemecek, T., & Schnetzer, J. (2011). *Methods of assessment of direct field emissions for LCIs of agricultural production systems. Data v3.0 (2012)*. Agroscope Reckenholz-Tänikon.

Neuweiler, R., & Krauss, J. (2017). *10/ Düngung im Gemüsebau. Grundlagen für die Düngung*

landwirtschaftlicher Kulturen in der Schweiz (GRUD).

Phocaides, A. (2007). *Handbook on pressurized irrigation techniques*. Food and Agriculture Organization of the United Nations.

Romero-Gámez, M., & Suárez-Rey, E. M. (2020). Environmental footprint of cultivating strawberry in Spain. *The International Journal of Life Cycle Assessment*, 25(4), 719–732. <https://doi.org/10.1007/s11367-020-01740-w>

Schenkel, P., Neumeister, L., & Greenpeace. (2016). *Pestizidcocktail in Schweizer Erdbeeren*. Greenpeace Schweiz.

Schweizer Obstverband. (2021). *Schweizer Obstverband, Jahresbericht 2020*.

Schweizer Obstverband. (2022, April 13). *Eträge Schweizer Erdbeeren nach Anbauart* [Persönliche Kommunikation].

Soode, E., Lampert, P., Weber-Blaschke, G., & Richter, K. (2015). Carbon footprints of the horticultural products strawberries, asparagus, roses and orchids in Germany. *Journal of Cleaner Production*, 87, 168–179. <https://doi.org/10.1016/j.jclepro.2014.09.035>

Stoessel, F., Juraske, R., Pfister, S., & Hellweg, S. (2012). Life Cycle Inventory and Carbon and Water FoodPrint of Fruits and Vegetables: Application to a Swiss Retailer. *Environmental Science & Technology*, 46(6), 3253–3262. <https://doi.org/10.1021/es2030577>

Strickhof (Hrsg.). (2016). *Kulturblatt Spargel*.

Swiss Vegetable Producers Association (Hrsg.). (2022). *Anbau und Ernte Grün- und Bleichspargel Schweiz*. <https://www.gemuese.ch/schweizer-spargeln>

Theurl, M. C., Haberl, H., Erb, K.-H., & Lindenthal, T. (2014). Contrasted greenhouse gas emissions from local versus long-range tomato production. *Agronomy for Sustainable Development*, 34(3), 593–602. <https://doi.org/10.1007/s13593-013-0171-8>

Valiante, D., Sirtori, I., Cossa, S., Corengia, L., Pedretti, M., Cavallaro, L., Vignoli, L., Galvagni, A., Gomasasca, S., Pesce, G. R., Boccadelli, A., Orsi, L., Lovarelli, D., Facchinetti, D., Pessina, D., & Bacenetti, J. (2019). Environmental impact of strawberry production in Italy and Switzerland with different cultivation practices. *Science of The Total Environment*, 664, 249–261. <https://doi.org/10.1016/j.scitotenv.2019.02.046>

Vazquez-Rowe, I., Kahhat, R., Quispe, I., & Bentín, M. (2015). *Environmental profile of green asparagus production in a hyper-arid zone in coastal Peru [LCA]*.

A1 Global warming potential

The global warming potential correlates with the environmental footprint based on the ecological scarcity method for strawberries. For asparagus, Peruvian asparagus transported by ship shows a similar carbon footprint as Swiss asparagus because pesticide emissions are not accounted for.

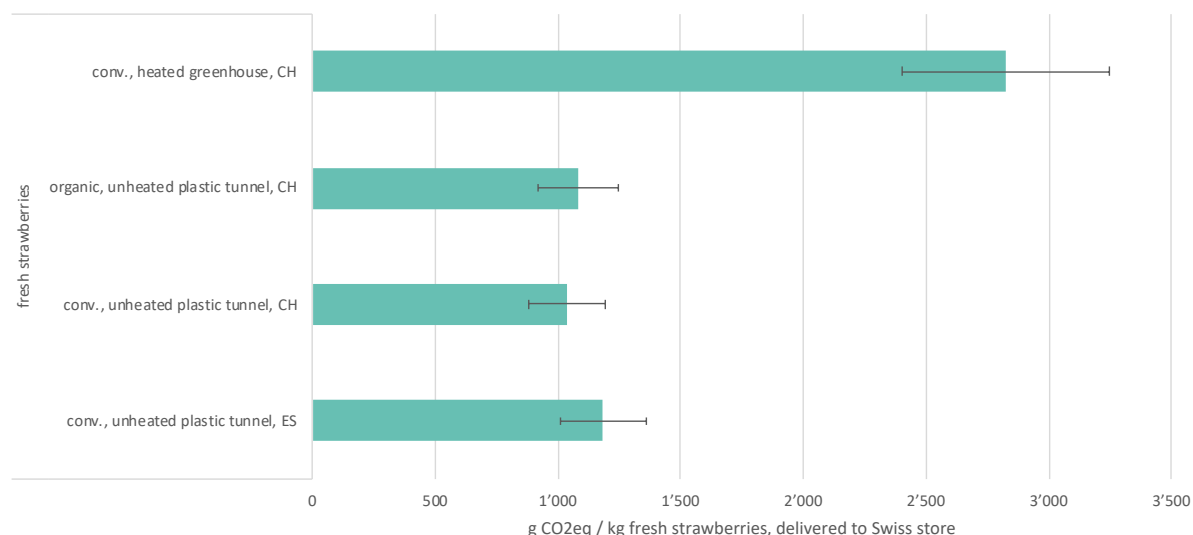


Figure 8: Carbon footprint for strawberries, delivered to Swiss store

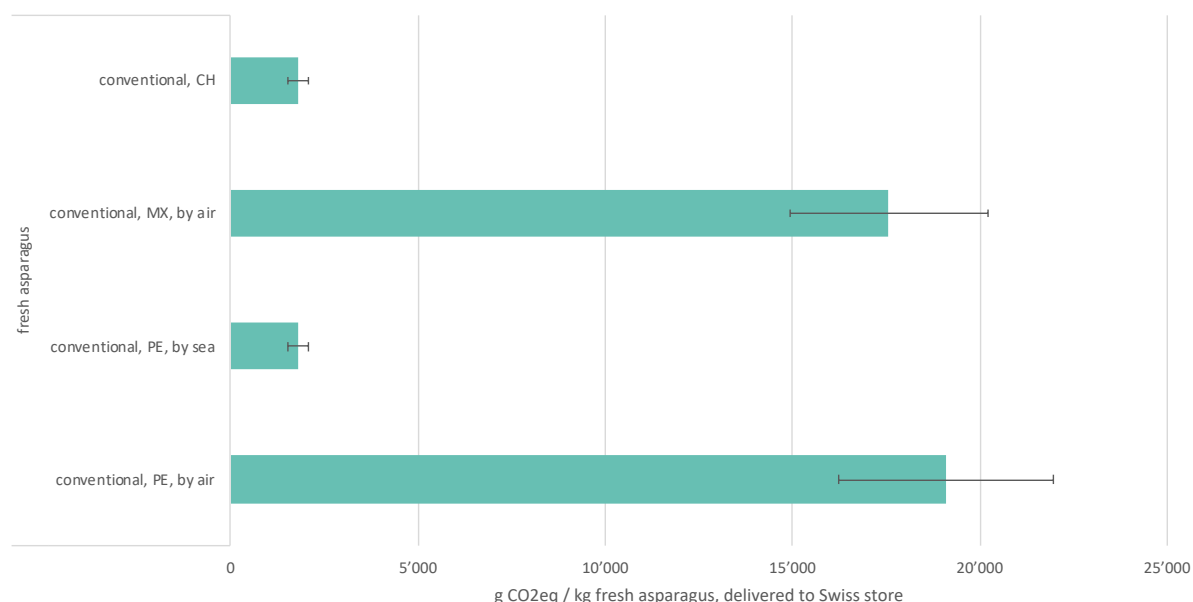


Figure 9: Carbon footprint for asparagus, delivered to Swiss store

A2 EF v3.0 method

The environmental footprint according to the EF v3.0 methods does not correlate well with the environmental footprint based on the ecological scarcity method.

For strawberry production the difference between heated and unheated strawberry production is less clear due to higher importance of water scarcity (ES production) and acidification and eutrophication (organic CH) in the EF method compared to the ecological scarcity method.

For asparagus production, the Peruvian asparagus transported by ship shows a lower impact than Mexican asparagus imported by air because of lower importance of the pesticide emissions.

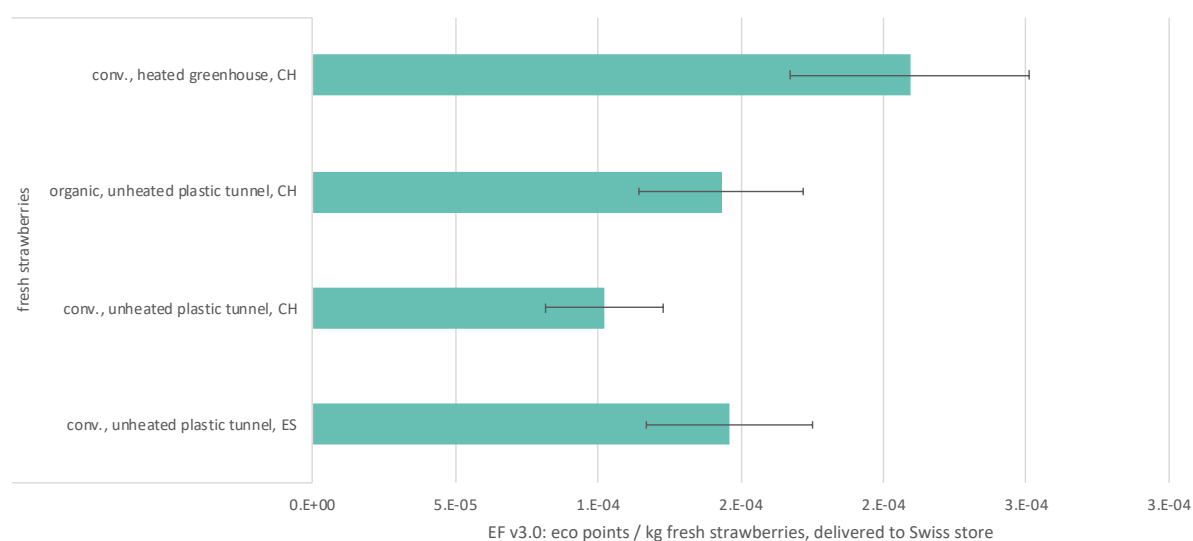


Figure 10: Environmental footprint according to EF v3.0 for strawberries, delivered to Swiss store

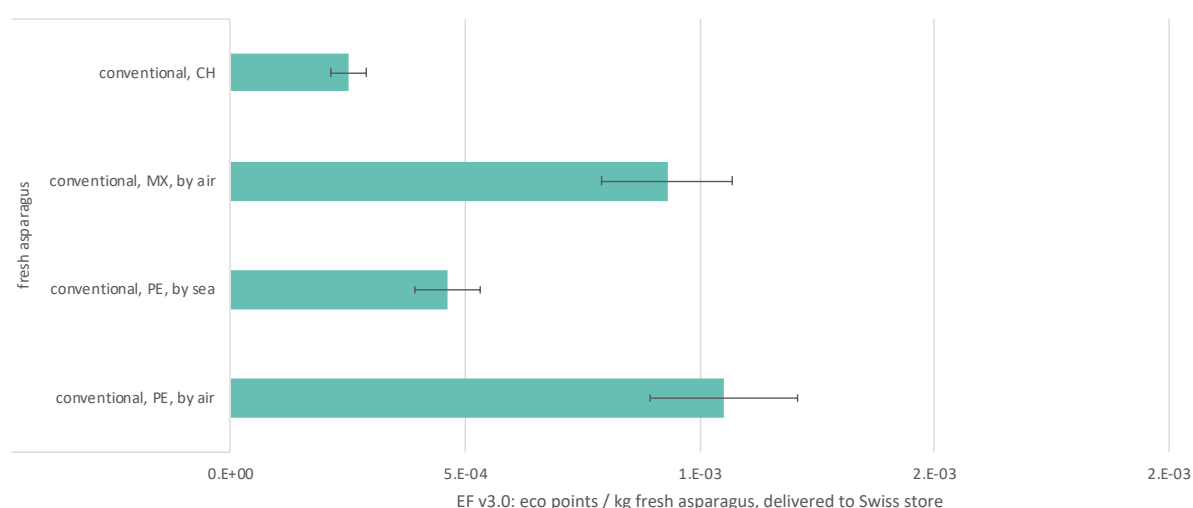


Figure 11: Environmental footprint according to EF v3.0 for asparagus, delivered to Swiss store

A3 Table with results

Table 15: Results for strawberries calculated with ecological scarcity 2021

eco points / kg strawberries, delivered Swiss store	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated plastic tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated multi tunnel/ES
Fertilisers (production and field emissions)	180	337	527	181
Pesticides (production and field emissions)	70	251	0	30
Water scarcity	0	1	1	216
Further cultivation processes	63	365	385	201
Tunnel / greenhouse infrastructure	1'913	1'025	1'196	1'007
Heating	1'931	0	0	0
Packaging	414	414	414	414
Transportation	66	66	66	456
Total	4'636	2'460	2'589	2'505

Table 16: Results for asparagus calculated with ecological scarcity 2021

eco points / kg asparagus, delivered Swiss store	Asparagus, conventional/CH	Asparagus, conventional/PE, by air	Asparagus, conventional/PE, by sea	Asparagus, conventional/MX, by air
Fertilisers (production and field emissions)	1'739	1'526	1'526	1'703
Pesticides (production and field emissions)	136	24'162	24'162	206
Water scarcity	0	435	435	583
Further cultivation processes	2'153	766	766	1'194
Packaging	525	893	893	893
Transportation	66	20'538	515	18'323
Total	4'618	48'320	28'297	22'902

Table 17: Carbon footprint of strawberries

g CO ₂ eq / kg strawberries, delivered Swiss store	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated plastic tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated multi tunnel/ES
Total	2'825	1'039	1'083	1'185

Table 18: Carbon footprint of asparagus

g CO ₂ eq / kg asparagus, delivered Swiss store	Asparagus, conventional/CH	Asparagus, conventional/PE, by air	Asparagus, conventional/PE, by sea	Asparagus, conventional/MX, by air
Total	1'785	19'096	1'788	17'560

Table 19: Results for strawberries calculated with EF v3.0

Eco points / kg strawberries, delivered Swiss store	Strawberries, conventional, heated greenhouse/CH	Strawberries, conventional, unheated plastic tunnel/CH	Strawberries, organic, unheated plastic tunnel/CH	Strawberries, conventional, unheated multi tunnel/ES
Total	2.09E-04	1.02E-04	1.43E-04	1.46E-04
Climate change	7.51E-05	2.77E-05	3.46E-05	2.57E-05
Ozone depletion	3.19E-07	2.80E-08	2.86E-08	4.60E-08
Ionising radiation	1.27E-06	5.02E-07	5.80E-07	6.32E-07
Photochemical ozone formation	4.85E-06	2.74E-06	3.30E-06	2.98E-06
Particulate Matter	1.28E-05	8.10E-06	2.05E-05	7.37E-06
Human toxicity, non-cancer	2.78E-06	1.54E-06	1.70E-06	2.29E-06
Human toxicity, cancer	2.67E-06	1.03E-06	1.40E-06	1.35E-06
Acidification terrestrial and freshwater	9.43E-06	5.05E-06	1.67E-05	4.82E-06
Eutrophication, freshwater	1.50E-06	1.18E-06	1.78E-06	7.70E-07
Eutrophication, marine	2.94E-06	2.98E-06	5.32E-06	1.67E-06
Eutrophication, terrestrial	4.75E-06	2.80E-06	1.24E-05	2.28E-06
Ecotoxicity, freshwater	2.06E-05	2.35E-05	1.45E-05	1.25E-05
Land use	2.80E-06	2.67E-06	2.63E-06	2.63E-06
Water use	1.83E-06	1.43E-06	1.45E-06	5.64E-05
Resource use, fossils	5.21E-05	1.87E-05	2.36E-05	1.81E-05
Resource use, minerals and metals	1.36E-05	2.20E-06	2.76E-06	6.31E-06

Table 20: Results for asparagus calculated with EF v3.0

Eco points / kg strawberries, delivered Swiss store	Asparagus, conventional/CH	Asparagus, conventional/PE, by air	Asparagus, conventional/PE, by sea	Asparagus, conventional/MX, by air
Total	2.50E-04	1.05E-03	4.62E-04	9.30E-04
Climate change	4.75E-05	3.69E-04	4.79E-05	3.43E-04
Ozone depletion	1.08E-07	2.27E-07	1.09E-07	1.92E-07
Ionising radiation	1.70E-06	1.36E-06	7.01E-07	1.32E-06
Photochemical ozone formation	9.76E-06	3.87E-05	6.50E-06	3.58E-05
Particulate Matter	1.60E-05	2.24E-05	1.55E-05	2.55E-05
Human toxicity, non-cancer	4.02E-06	4.81E-06	3.11E-06	3.62E-06
Human toxicity, cancer	1.58E-06	1.53E-06	7.30E-07	1.76E-06
Acidification terrestrial and freshwater	1.61E-05	5.04E-05	1.69E-05	5.12E-05
Eutrophication, freshwater	5.29E-06	3.66E-06	3.18E-06	3.97E-06
Eutrophication, marine	1.62E-05	2.70E-05	1.17E-05	2.71E-05
Eutrophication, terrestrial	1.17E-05	3.27E-05	1.01E-05	3.06E-05
Ecotoxicity, freshwater	4.37E-05	1.95E-04	1.59E-04	6.05E-05
Land use	3.56E-05	2.50E-05	2.44E-05	1.07E-05
Water use	3.42E-06	1.20E-04	1.29E-04	1.83E-04
Resource use, fossils	3.26E-05	1.54E-04	2.88E-05	1.46E-04
Resource use, minerals and metals	4.92E-06	4.99E-06	4.32E-06	5.21E-06

A4 Metadata and unit process raw data

The life cycle inventories for the newly modelled and updated processes are provided as unit process raw data in the EcoSpold v1 format (unit process in SimaPro). The electronic data includes the full EcoSpold v1 documentation.

For each investigated process, two types of tables (X-Process and X-Exchange) are provided. For better readability of the whole report, these data are listed here in the Annex.

A4.1 Auxiliary inventories

ReferenceFunction	Name	strawberry seedling, for planting, in unheated greenhouse
Geography	Location	RER
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	p
	IncludedProcesses	The inventory includes the processes applied in this phase, that is the processes of irrigating, fertilisation, pest and pathogen control, harvesting; the greenhouse and the machines and shed or surface used to park them; all inputs as seeds, fertilizers (mineral and organic), active substances, water for irrigation, fuels and energy as well as the transport to the farm.
	Amount	1
	LocalName	Erdbeersetzling, zum Einpflanzen, in unbeheiztem Gewächshaus
	Synonyms	
	GeneralComment	This inventory refers to 1 strawberry seedling.
	InfrastructureIncluded	1
	Category	agricultural means of production
	SubCategory	seed
	LocalCategory	Landwirtschaftliche Produktionsmittel
	LocalSubCategory	Saatgut
	Formula	
	StatisticalClassification	
	CASNumber	
TimePeriod	StartDate	2017
	EndDate	2020
	DataValidForEntirePeriod	1
	OtherPeriodText	
Geography	Text	The inventory is modelled for Europe
Technology	Text	
Representativeness	Percent	
	ProductionVolume	
	SamplingProcedure	The data is based on literature and expert judgements.
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 1: Metadata of strawberry seedling, for planting, in unheated greenhouse/RER

	Name	Location	Infrastructure Process	Unit	strawberry seedling, for planting, in unheated greenhouse	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				RER			
	Infrastructure Process				0			
	Unit				p			
product	strawberry seedling, for planting, in unheated greenhouse	RER	0	p	1.0	0		
resource, in water	Water, river	-	-	m3	5.10E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
resource, biotic	Peat	-	-	kg	3.00E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
technosphere	plastic tunnel	ES	1	m2a	7.69E-07	1.00E+00	3.38E+00 (4,4,4,2,4,5,BU:3); ;	
	installation for drip irrigation	CH	0	m3	5.10E-04	1.00E+00	3.38E+00 (4,4,4,2,4,5,BU:3); ;	
	electricity, low voltage, production							
	ENTSO, at grid	ENTSO	0	kWh	1.22E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	average mineral fertiliser, as P2O5, at regional storehouse	RER	0	kg	4.19E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	average mineral fertiliser, as N, at regional storehouse	RER	0	kg	3.28E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	average mineral fertiliser, as K2O, at regional storehouse	RER	0	kg	1.63E-03	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	pesticide unspecified, at regional storehouse	RER	0	kg	2.88E-08	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	copper oxide, at plant	RER	0	kg	1.15E-06	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
emission air, low population density	Water	-	-	kg	0.04.59E-7	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
	Ammonia	-	-	kg	1.27E-05	1.00E+00	1.74E+00 (4,4,4,2,4,5,BU:1.2); ;	
	Dinitrogen monoxide	-	-	kg	9.72E-06	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
	Nitrogen oxides	-	-	kg	1.53E-05	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
emission water, river	Water	-	-	kg	4.59E-07	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
emission water, ground-	Water	-	-	kg	1.02E-05	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
emission soil, agricultural	Cadmium	-	-	kg	2.88E-08	1.00E+00	1.70E+00 (4,4,4,2,4,5,BU:1.1); ;	
	Copper	-	-	kg	1.15E-06	1.00E+00	1.70E+00 (4,4,4,2,4,5,BU:1.1); ;	

Figure 2: Unit process raw data of strawberry seedling, for planting, in unheated greenhouse/RER

ReferenceFunction	Name	strawberry seedling, organic, for planting, in unheated greenhouse
Geography	Location	RER
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	p
	IncludedProcesses	The inventory includes the processes applied in this phase, that is the processes of irrigating, fertilisation, pest and pathogen control, harvesting; the greenhouse and the machines and shed or surface used to park them; all inputs as seeds, fertilizers (mineral and organic), active substances, water for irrigation, fuels end energy as well as the transport to the farm.
	Amount	1
	LocalName	Erdbeersetzling, Bio, zum Einpflanzen, in unbeheiztem Gewächshaus
	Synonyms	
	GeneralComment	This inventory refers to 1 strawberry seedling.
	InfrastructureIncluded	1
	Category	agricultural means of production
	SubCategory	seed
	LocalCategory	Landwirtschaftliche Produktionsmittel
	LocalSubCategory	Saatgut
	Formula	
	StatisticalClassification	
	CASNumber	
TimePeriod	StartDate	2017
	EndDate	2020
	DataValidForEntirePeriod	1
	OtherPeriodText	
Geography	Text	The inventory is modelled for Europe
Technology	Text	
Representativeness	Percent	
	ProductionVolume	
	SamplingProcedure	The data is based on literature and expert judgements.
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 3: Metadata of strawberry seedling, organic, for planting, in unheated greenhouse/RER

	Name	Location	Infrastructure Process	Unit	strawberry seedling, organic, for planting, in unheated greenhouse	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				RER			
	Infrastructure Process				0			
	Unit				p			
product	strawberry seedling, organic, for planting, in unheated greenhouse	RER	0	p	1.0	0		
resource, in water	Water, river	-	-	m3	5.10E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
resource, biotic	Peat	-	-	kg	3.00E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
technosphere	plastic tunnel	ES	1	m2a	7.69E-07	1.00E+00	3.38E+00 (4,4,4,2,4,5,BU:3); ;	
	installation for drip irrigation	CH	0	m3	5.10E-04	1.00E+00	3.38E+00 (4,4,4,2,4,5,BU:3); ;	
	electricity, low voltage, production	ENTSO	0	kWh	1.22E-04	1.00E+00	1.69E+00 (4,4,4,2,4,5,BU:1.05); ;	
	ENTSO, at grid							
emission air, low population density	Water	-	-	kg	0.04.59E-7	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
	Ammonia	-	-	kg	1.27E-05	1.00E+00	1.74E+00 (4,4,4,2,4,5,BU:1.2); ;	
	Dinitrogen monoxide	-	-	kg	9.72E-06	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
	Nitrogen oxides	-	-	kg	1.53E-05	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
emission water, river	Water	-	-	kg	4.59E-07	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	
emission water, ground-	Water	-	-	kg	1.02E-05	1.00E+00	1.94E+00 (4,4,4,2,4,5,BU:1.5); ;	

Figure 4: Unit process raw data of strawberry seedling, organic, for planting, in unheated greenhouse/RER

ReferenceFunction	Name	asparagus seedling, open field, at production site
Geography	Location	GLO
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	p
TimePeriod	IncludedProcesses	The inventory includes the irrigating, fertilisation, pest and pathogen control; all inputs as fertilizers (mineral and organic), active substances, water for irrigation, fuels end energy as well as the transport to the farm.
	LocalName	Spargelsetzling, offenes Feld, ab Produktionsort
	Synonyms	
	GeneralComment	This inventory refers to 1 asparagus seedling.
	InfrastructureIncluded	1
	Category	agricultural means of production
	SubCategory	seed
	LocalCategory	Landwirtschaftliche Produktionsmittel
	LocalSubCategory	Saatgut
	Formula	
	StatisticalClassification	
	CASNumber	
	StartDate	2015
	EndDate	2022
Geography	DataValidForEntirePeriod	1
	OtherPeriodText	
	Text	The inventory is globally modelled.
	Text	
	Percent	
	ProductionVolume	
	SamplingProcedure	The data is based on literature and expert judgements.
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 5: Metadata of asparagus seedling, open field, at production site/GLO

	Name	Location	Infrastructure Process	Unit	asparagus seedling, open field, at production site	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				GLO			
	Infrastructure Process				0			
	Unit				p			
product	asparagus seedling, open field, at production site	GLO	0	p	1.0	0		
resource, land	Transformation, from annual crop, non-irrigated	-	-	m2	4.00E-02	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, to annual crop, non-irrigated	-	-	m2	4.00E-02	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Occupation, annual crop	-	-	m2a	4.00E-02	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
technosphere	diesel, burned in agricultural machine	CH	0	kg	8.24E-05	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	poultry manure, dried, at regional storehouse	CH	0	kg	1.60E-04	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	ammonium nitrate, as N, at regional storehouse	RER	0	kg	3.48E-04	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	phosphoric acid, fertiliser grade, 70% in H2O, at plant	GLO	0	kg	4.39E-04	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	potassium sulphate, as K2O, at regional storehouse	RER	0	kg	2.32E-04	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	magnesium sulphate, at plant	RER	0	kg	8.00E-06	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	calcium nitrate, as N, at regional storehouse	RER	0	kg	8.00E-06	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	potassium nitrate, as N, at regional storehouse	RER	0	kg	1.20E-04	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	installation for drip irrigation	CH	0	m3	1.34E-02	1.00E+00	3.06E+00	(1,2,3,3,1,5,BU:3); ;
	tap water, unspecified natural origin OECD, at user	RER	0	kg	1.34E-02	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	pesticide unspecified, at regional storehouse	RER	0	kg	3.35E-06	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
	transport, freight, lorry 32-40 metric ton, EURO 4	RER	0	tkm	1.45E-03	1.00E+00	2.11E+00	(4,2,3,3,1,5,BU:2); ;
	transport, freight, lorry 32-40 metric ton, EURO 4	RER	0	tkm	1.54E-02	1.00E+00	2.11E+00	(4,2,3,3,1,5,BU:2); ;
	transport, aircraft, freight, long-haul	RER	0	tkm	9.21E-03	1.00E+00	2.11E+00	(4,2,3,3,1,5,BU:2); ;
	transport, tractor and trailer	CH	0	tkm	8.52E-04	1.00E+00	2.11E+00	(4,2,3,3,1,5,BU:2); ;
	transport, transoceanic freight ship	OCE	0	tkm	2.54E-03	1.00E+00	2.11E+00	(4,2,3,3,1,5,BU:2); ;
	electricity, low voltage, production GLO, at grid	GLO	0	kWh	6.40E-03	1.00E+00	1.24E+00	(1,2,3,3,1,5,BU:1.05); ;
emission air, low population density	Ammonia	-	-	kg	1.70E-05	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2); ;
	Nitrogen oxides	-	-	kg	1.92E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
	Dinitrogen monoxide	-	-	kg	9.20E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
	Water, OECD	-	-	kg	7.26E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
emission water, ground-	Nitrate	-	-	kg	6.16E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
	Phosphorus	-	-	kg	2.80E-07	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
emission water, river	Phosphorus	-	-	kg	2.96E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
	Water	-	-	kg	7.26E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5); ;
	Cadmium, ion	-	-	kg	8.80E-12	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3); ;
	Copper, ion	-	-	kg	8.80E-09	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3); ;
	Zinc, ion	-	-	kg	3.36E-08	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5); ;
	Lead	-	-	kg	1.44E-10	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5); ;
	Nickel, ion	-	-	kg	2.80E-11	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5); ;
	Chromium, ion	-	-	kg	2.88E-08	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3); ;
	Mercury	-	-	kg	3.64E-13	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5); ;
emission soil, agricultural	Imidacloprid	-	-	kg	3.38E-07	1.00E+00	1.32E+00	(1,2,3,3,1,5,BU:1.2); ;
	Methomyl	-	-	kg	2.16E-06	1.00E+00	1.32E+00	(1,2,3,3,1,5,BU:1.2); ;
	Sulfur	-	-	kg	7.60E-08	1.00E+00	1.58E+00	(1,2,3,3,1,5,BU:1.5); ;
	Difenoconazole	-	-	kg	4.80E-07	1.00E+00	1.32E+00	(1,2,3,3,1,5,BU:1.2); ;
	Deltamethrin	-	-	kg	3.00E-04	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2); ;
	Mercury	-	-	kg	1.44E-11	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1); ;

Figure 6: Unit process raw data of asparagus seedling, open field, at production site/GLO

ReferenceFunction	Name	installation for drip irrigation
Geography	Location	CH
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	m3
TimePeriod	IncludedProcesses	This inventory includes the water pumping, energy use, infrastructure, but not the water emissions of irrigation (shall be included in the crop production dataset).
	LocalName	Installation Tropfsystem Bewässerung
	Synonyms	
	GeneralComment	This dataset represents the drip irrigation in Switzerland with a 90% irrigation efficiency. The inventory includes the water pumping of ground water and surface water, as well as the infrastructure. The water supplied to the plants (1m3) and the water evaporated or infiltrated shall be included in the crop production datasets.
	InfrastructureIncluded	1
	Category	water supply
	SubCategory	production
	LocalCategory	Wasserversorgung
	LocalSubCategory	Bereitstellung
	Formula	
	StatisticalClassification	
	CASNumber	
	StartDate	2015
	EndDate	2020
	DataValidForEntirePeriod	1
Geography	OtherPeriodText	
	Text	The inventory is modelled for Switzerland
	Text	
	Percent	
	ProductionVolume	
Technology	SamplingProcedure	Literature values and estimations.
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 7: Metadata of installation for drip irrigation/CH

	Name	Location	Infrastructure Process	Unit	installation for drip irrigation	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				CH			
	Infrastructure Process				0			
	Unit				m3			
product	installation for drip irrigation	CH	0	m3	1.0	0		
technosphere	brass, at plant	CH	0	kg	6.20E-04	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	cast iron, at plant	RER	0	kg	1.80E-05	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	extrusion, plastic pipes	RER	0	kg	1.07E-02	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	injection moulding	RER	0	kg	9.90E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	steel product manufacturing, average metal working	RER	0	kg	1.09E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	injection moulding	RER	0	kg	9.90E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	nylon 66, at plant	RER	0	kg	3.10E-06	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	polyethylene, HDPE, granulate, at plant	RER	0	kg	2.60E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	polyethylene, LDPE, granulate, at plant	RER	0	kg	8.10E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	polypropylene, granulate, at plant	RER	0	kg	9.90E-03	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	steel, low-alloyed, at plant	RER	0	kg	4.50E-04	1.00E+00	1.42E+00	(3,4,4,3,3,5,BU:1.05); ;
	diesel, burned in building machine, average	CH	0	MJ	3.94E-05	1.00E+00	2.20E+00	(3,5,4,3,3,5,BU:2); ;
	disposal, inert material, 0% water, to sanitary landfill	CH	0	kg	3.10E-06	1.00E+00	1.52E+00	(4,5,4,5,3,5,BU:1.05); ;
	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.07E-02	1.00E+00	1.52E+00	(4,5,4,5,3,5,BU:1.05); ;
	disposal, polypropylene, 15.9% water, to municipal incineration	CH	0	kg	9.90E-03	1.00E+00	1.52E+00	(4,5,4,5,3,5,BU:1.05); ;
	disposal, building, bulk iron (excluding reinforcement), to sorting plant	CH	0	kg	1.09E-03	1.00E+00	1.52E+00	(4,5,4,5,3,5,BU:1.05); ;

Figure 8: Unit process raw data of installation for drip irrigation/CH

ReferenceFunction	Name	diesel, burned in agricultural machine
Geography	Location	CH
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
	IncludedProcesses	The inventory includes the diesel demand and the combustions emissions as well as the agricultural machinery
	LocalName	Diesel, verbrannt in landwirtschaftlichen Maschinen
	Synonyms	
	GeneralComment	This dataset represents the diesel demand in agricultural machinery and is valid for 1kg diesel, burned in agricultural machinery
	InfrastructureIncluded	1
	Category	agricultural production
	SubCategory	fuels
	LocalCategory	Landwirtschaftliche Produktion
	LocalSubCategory	Brenn- und Treibstoffe
	Formula	
	StatisticalClassification	
	CASNumber	
TimePeriod	StartDate	2014
	EndDate	2022
	DataValidForEntirePeriod	1
	OtherPeriodText	
Geography	Text	The inventory is modelled for Europe
Technology	Text	
Representativeness	Percent	
	ProductionVolume	
	SamplingProcedure	The data is based on literature and expert judgements.
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 9: Metadata of diesel, burned in agricultural machine/CH

	Name	Location	Infrastructure Process	Unit	diesel, burned in agricultural machine	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				CH			
	Infrastructure Process				0			
	Unit				kg			
product	diesel, burned in agricultural machine	CH	0	kg	1.0	0		
technosphere	tractor, production	CH	1	kg	8.37E-02	1.00E+00	3.06E+00	(1,2,3,2,1,5,BU:3); ;
	trailer, production	CH	1	kg	5.55E-01	1.00E+00	3.06E+00	(1,2,3,2,1,5,BU:3); ;
	diesel, at regional storage	CH	0	kg	1.00E+00	1.00E+00	1.30E+00	(1,2,4,2,1,5,BU:1.05); ;
	shed	CH	1	m2	1.86E-03	1.00E+00	3.09E+00	(1,2,4,2,1,5,BU:3); ;
emission air, low population density	NMVOG, non-methane volatile organic compounds, unspecified origin	-	-	kg	2.14E-03	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
	Nitrogen oxides	-	-	kg	3.90E-02	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
	Carbon monoxide, fossil	-	-	kg	5.87E-03	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Carbon dioxide, fossil	-	-	kg	3.12E+00	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Sulfur dioxide	-	-	kg	1.01E-03	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Methane, fossil	-	-	kg	1.29E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Benzene	-	-	kg	7.29E-06	1.00E+00	3.05E+00	(2,2,2,1,1,5,BU:3);
	Particulates, < 2.5 um	-	-	kg	4.91E-03	1.00E+00	3.05E+00	(2,2,2,1,1,5,BU:3);
	Cadmium	-	-	kg	1.00E-08	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Chromium	-	-	kg	5.00E-08	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Copper	-	-	kg	1.70E-06	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Dinitrogen monoxide	-	-	kg	1.20E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Nickel	-	-	kg	7.00E-08	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Zinc	-	-	kg	1.00E-06	1.00E+00	5.06E+00	(2,2,2,1,1,5,BU:5);
	Benzo(a)pyrene	-	-	kg	3.00E-08	1.00E+00	3.05E+00	(2,2,2,1,1,5,BU:3);
	PAH, polycyclic aromatic hydrocarbons	-	-	kg	3.28E-06	1.00E+00	3.05E+00	(2,2,2,1,1,5,BU:3);
	Heat, waste	-	-	MJ	4.54E+01	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Ammonia	-	-	kg	2.00E-05	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Selenium	-	-	kg	1.00E-08	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
emission soil, agricultural	Zinc	-	-	kg	4.82E-04	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1);
	Lead	-	-	kg	7.87E-07	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1);
	Cadmium	-	-	kg	1.79E-07	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1);

Figure 10: Unit process raw data of diesel, burned in agricultural machine/CH

A4.2 Cultivation

ReferenceFunction	Name	strawberries, conventional, heated greenhouse
Geography	Location	CH
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
	IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, heating, harvesting; the greenhouse and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as transports to the farm and the direct field emissions.
	LocalName	Erdbeeren, konventionell, beheiztes Gewächshaus
	Synonyms	
	GeneralComment	The inventory refers to the production of 1 kg of conventional strawberries in a heated greenhouse in Switzerland.
	InfrastructureIncluded	1
	Category	agricultural production
	SubCategory	plant production
	LocalCategory	Landwirtschaftliche Produktion
	LocalSubCategory	Pflanzenbau
	Formula	
	StatisticalClassification	
	CASNumber	
TimePeriod	StartDate	2016
	EndDate	2022
	DataValidForEntirePeriod	1
	OtherPeriodText	
Geography	Text	The inventory is modelled for Switzerland
Technology	Text	
Representativeness	Percent	
	ProductionVolume	
	SamplingProcedure	Literature and data from producer
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 11: Metadata of strawberries, conventional, heated greenhouse/CH

	Name	Location	Infrastructure Process	Unit	strawberries, conventional, heated greenhouse	Uncertainty Type	Standard Deviation 95%	General Comment
	Location Infrastructure Process Unit				CH 0 kg 1.0	0		
product	strawberries, conventional, heated greenhouse	CH	0	kg				
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.83E-01	1.00E+00	1.22E+00	(2,1,1,1,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.94E+00	1.00E+00	1.22E+00	(2,1,1,1,1,5,BU:1.05);
technosphere	average mineral fertiliser, as N, at regional storehouse	RER	0	kg	4.23E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	magnesium sulphate, at plant	RER	0	kg	9.62E-04	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	average mineral fertiliser, as P ₂ O ₅ , at regional storehouse	RER	0	kg	2.00E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	average mineral fertiliser, as K ₂ O, at regional storehouse	RER	0	kg	4.75E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	strawberry seedling, for planting, in unheated greenhouse	RER	0	p	8.65E-01	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	tap water, unspecified natural origin CH, at user	CH	0	kg	1.58E-02	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	polypropylene, granulate, at plant	RER	0	kg	1.92E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	extrusion, plastic film	RER	0	kg	1.92E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	greenhouse, glass walls and roof, metal tubes	CH	1	m ² a	1.92E-01	1.00E+00	3.05E+00	(2,3,1,1,1,5,BU:3);
	pesticide unspecified, at regional storehouse	RER	0	kg	3.06E-05	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	transport, freight, lorry, fleet average	CH	0	tkm	4.17E-03	1.00E+00	2.05E+00	(2,3,1,1,1,5,BU:2);
	heat, natural gas, at boiler modulating >100kW	RER	0	MJ	2.22E+01	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	electricity, low voltage, at grid	CH	0	kWh	6.15E-02	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	electricity, low voltage, at grid	CH	0	kWh	3.64E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.92E-03	1.00E+00	1.22E+00	(2,3,1,1,1,5,BU:1.05);
emission air, low population density	Dinitrogen monoxide	-	-	kg	7.31E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Ammonia	-	-	kg	2.21E-04	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Carbon dioxide, fossil	-	-	kg	0.00E+00	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);
	Nitrogen oxides	-	-	kg	1.54E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission air, unspecified	Water, CH	-	-	kg	1.43E+01	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, unspecified	Water, CH	-	-	m ³	6.38E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	2.48E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	1.35E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Phosphorus	-	-	kg	1.44E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Cadmium, ion	-	-	kg	8.65E-10	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Copper, ion	-	-	kg	6.15E-08	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Zinc, ion	-	-	kg	3.08E-07	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Lead	-	-	kg	2.31E-09	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Nickel, ion	-	-	kg	1.50E-10	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Chromium, ion	-	-	kg	3.65E-07	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
emission soil, agricultural	Azoxystrobin	-	-	kg	3.85E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Bupirimate	-	-	kg	7.69E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Cyprodinil	-	-	kg	4.42E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Difenoconazole	-	-	kg	7.69E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Etoazazole	-	-	kg	1.92E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	N-(2,3-dichloro-4-hydroxyphenyl)-1-methylcyclohexane-1-carboxamide	-	-	kg	1.02E-05	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Fludioxonil	-	-	kg	1.92E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Zoxamide	-	-	kg	1.54E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Kresoxim-methyl	-	-	kg	1.73E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Myclobutanil	-	-	kg	3.85E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Phenmedipham	-	-	kg	2.31E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Quinoxifen	-	-	kg	1.92E-07	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Thiacloprid	-	-	kg	2.50E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Trifloxystrobin	-	-	kg	3.08E-06	1.00E+00	1.37E+00	(4,1,1,1,1,5,BU:1.2);
	Cadmium	-	-	kg	6.46E-08	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1);
emission economic-issue, unspecified	Water, embodied in product, CH	-	-	kg	9.00E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 12: Unit process raw data of strawberries, conventional, heated greenhouse/CH

ReferenceFunction	Name	strawberries, conventional, unheated plastic tunnel
Geography	Location	CH
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
TimePeriod	IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting; the tunnel and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as all transports to the farm and the direct field emissions.
	LocalName	Erdbeeren, konventionell, unbeheizter Folientunnel
	Synonyms	
	GeneralComment	The inventory refers to the production of 1 kg of conventional strawberries in a unheated double layer tunnel in Switzerland.
	InfrastructureIncluded	1
	Category	agricultural production
	SubCategory	plant production
	LocalCategory	Landwirtschaftliche Produktion
	LocalSubCategory	Pflanzenbau
	Formula	
	StatisticalClassification	
	CASNumber	
	StartDate	2016
	EndDate	2022
	DataValidForEntirePeriod	1
	OtherPeriodText	
Geography	Text	The inventory is modelled for Switzerland
Technology	Text	
Representativeness	Percent	
	ProductionVolume	
	SamplingProcedure	Based on literature and producer interviews
	Extrapolations	none
	UncertaintyAdjustments	none

Figure 13: Metadata of strawberries, conventional, unheated plastic tunnel/CH

	Name	Location	Infrastructure Process	Unit	strawberries, conventional, unheated plastic tunnel	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				CH			
	Infrastructure Process				0			
	Unit				kg			
product	strawberries, conventional, unheated plastic tunnel	CH	0	kg	1.0	0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.83E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.94E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
technosphere	average mineral fertiliser, as N, at regional storehouse	RER	0	kg	7.14E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	average mineral fertiliser, as K ₂ O, at regional storehouse	RER	0	kg	8.64E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	average mineral fertiliser, as P ₂ O ₅ , at regional storehouse	RER	0	kg	2.43E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	magnesium sulphate, at plant	RER	0	kg	1.43E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	pesticide unspecified, at regional storehouse	CH	0	kg	1.14E-04	1.00E+00	1.24E+00	(2,1,3,1,1,5,BU:1.05);
	tap water, unspecified natural origin CH, at user	CH	0	kg	3.52E+01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
	plastic tunnel	CH	1	m ² a	7.14E-01	1.00E+00	3.06E+00	(3,3,1,1,1,5,BU:3);
	strawberry seedling, for planting, in unheated greenhouse	RER	0	p	3.21E+00	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.29E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	polyethylene, LDPE, granulate, at plant	RER	0	kg	3.50E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	extrusion, plastic pipes	RER	0	kg	1.29E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	extrusion, plastic film	RER	0	kg	3.50E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	diesel, burned in agricultural machine	CH	0	kg	3.18E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	transport, freight, lorry, fleet average	CH	0	tkm	1.66E-02	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
	electricity, low voltage, production CH, at grid	CH	0	kWh	8.14E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.81E+01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);
emission air, low regulation density	Dinitrogen monoxide	-	-	kg	1.40E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	3.74E-04	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Nitrogen oxides	-	-	kg	2.93E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission air, unspecified	Water, CH	-	-	kg	3.18E+01	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	5.36E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	5.00E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Phosphorus	-	-	kg	3.57E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Water, CH	-	-	m ³	4.93E+02	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
	Zinc, ion	-	-	kg	7.14E-07	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Chromium, ion	-	-	kg	1.43E-06	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
emission soil, agricultural	Azoxystrobin	-	-	kg	1.43E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Bupirimate	-	-	kg	2.86E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Cyprodinil	-	-	kg	1.64E-05	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Difenoconazole	-	-	kg	2.86E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Etozazole	-	-	kg	7.14E-07	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	N-(2,3-dichloro-4-hydroxyphenyl)-1-methylcyclohexane-1-carboxamide	-	-	kg	3.79E-05	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Fludioxonil	-	-	kg	7.14E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Zoxamide	-	-	kg	7.14E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Kresoxim-methyl	-	-	kg	6.43E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Myclobutanil	-	-	kg	1.43E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Phenmedipham	-	-	kg	8.57E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Quinoxifen	-	-	kg	7.14E-07	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Thiacloprid	-	-	kg	9.29E-06	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
	Trifloxystrobin	-	-	kg	1.14E-05	1.00E+00	1.32E+00	(2,1,3,1,1,5,BU:1.2);
emission economic-issue, unspecified	Water, embodied in product, CH	-	-	kg	9.00E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 14: Unit process raw data of strawberries, conventional, unheated plastic tunnel/CH

ReferenceFunction	Name	strawberries, organic, unheated plastic tunnel
Geography	Location	CH
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
TimePeriod	IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting; the tunnel and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as transports to the farm and the direct field emissions.
	LocalName	Erdbeeren Bio, unbeheizter Folientunnel
	Synonyms	
	GeneralComment	The inventory refers to the production of 1 kg of organic strawberries in a unheated double layer tunnel in Switzerland.
	InfrastructureIncluded	1
	Category	agricultural production
	SubCategory	plant production
	LocalCategory	Landwirtschaftliche Produktion
	LocalSubCategory	Pflanzenbau
	Formula	
	StatisticalClassification	
	CASNumber	
	StartDate	2017
	EndDate	2022
Geography	DataValidForEntirePeriod	1
	OtherPeriodText	
	Text	The inventory is modelled for Switzerland
	Text	
	Percent	
	ProductionVolume	
	SamplingProcedure	Based on literature and producer interviews
Representativeness	Extrapolations	none
	UncertaintyAdjustments	none

Figure 15: Metadata of strawberries, organic, unheated plastic tunnel/CH

	Name	Location	Infrastructure Process	Unit	strawberries, organic, unheated plastic tunnel	Uncertainty Type	Standard Deviation 95%	General Comment
	Location Infrastructure Process Unit				CH 0 kg 1.0			
product	strawberries, organic, unheated plastic tunnel	CH	0	kg		0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.83E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.94E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
technosphere	tap water, unspecified natural origin CH, at user	CH	0	kg	4.11E+01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
	plastic tunnel	CH	1	m2a	8.33E-01	1.00E+00	3.06E+00	(3,3,1,1,1,5,BU:3);
	strawberry seedling, organic, for planting, in unheated greenhouse	RER	0	p	3.75E+00	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	polyethylene, HDPE, granulate, at plant	RER	0	kg	1.50E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	polyethylene, LDPE, granulate, at plant	RER	0	kg	4.08E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	extrusion, plastic pipes	RER	0	kg	1.50E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	extrusion, plastic film	RER	0	kg	4.08E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	diesel, burned in agricultural machine	CH	0	kg	3.71E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	transport, freight, lorry, fleet average	CH	0	tkm	1.27E-02	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
	electricity, low voltage, production CH, at grid	CH	0	kWh	9.51E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
emission air, low population density	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.81E+01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);
	Dinitrogen monoxide	-	-	kg	8.41E-05	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	3.64E-03	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
emission air, unspecified	Nitrogen oxides	-	-	kg	1.77E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Water, CH	-	-	kg	3.71E+01	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	1.01E-02	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	5.83E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Phosphorus	-	-	kg	6.83E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Water, CH	-	-	m3	3.10E-03	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
	Cadmium, ion	-	-	kg	2.42E-09	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	Copper, ion	-	-	kg	3.00E-07	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	Zinc, ion	-	-	kg	2.42E-06	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Lead	-	-	kg	2.58E-08	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Nickel, ion	-	-	kg	1.00E-09	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Chromium, ion	-	-	kg	1.50E-06	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	Mercury	-	-	kg	1.08E-10	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
emission economic-issue, unspecified	Water, embodied in product, CH	-	-	kg	9.00E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 16: Unit process raw data of strawberries, organic, unheated plastic tunnel/CH

ReferenceFunction	Name	strawberries, conventional, unheated plastic tunnel
Geography	Location	ES
ReferenceFunction	InfrastructureProcess	0
ReferenceFunction	Unit	kg
TimePeriod	IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting; the tunnel and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as all transports to farm and the direct field emissions.
	LocalName	Erdbeeren, konventionell, unbeheizter Folientunnel
	Synonyms	
	GeneralComment	The inventory refers to the production of 1 kg of conventional strawberries in a unheated double layer tunnel in Spain.
	InfrastructureIncluded	1
	Category	agricultural production
	SubCategory	plant production
	LocalCategory	Landwirtschaftliche Produktion
	LocalSubCategory	Pflanzenbau
	Formula	
	StatisticalClassification	
	CASNumber	
	StartDate	2020
	EndDate	2022
Geography	DataValidForEntirePeriod	1
	OtherPeriodText	
	Text	The inventory is modelled for Spain.
	Text	
	Percent	
	ProductionVolume	
	SamplingProcedure	Based on literature and producer interviews
Representativeness	Extrapolations	none
	UncertaintyAdjustments	none

Figure 17: Metadata of strawberries, conventional, unheated plastic tunnel/ES

	Name	Location	Infrastructure Process	Unit	strawberries, conventional, unheated plastic tunnel	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				ES			
	Infrastructure Process				0			
	Unit				kg			
product	strawberries, conventional, unheated plastic tunnel	ES	0	kg	1.0	0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.83E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.94E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in water	Water, unspecified natural origin, ES	-	-	m3	1.17E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
technosphere	average mineral fertiliser, as N, at regional storehouse	RER	0	kg	5.00E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	average mineral fertiliser, as K2O, at regional storehouse	RER	0	kg	6.82E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	average mineral fertiliser, as P2O5, at regional storehouse	RER	0	kg	4.49E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	pesticide unspecified, at regional storehouse	RER	0	kg	1.08E-04	1.00E+00	1.22E+00	(2,1,1,1,1,5,BU:1.05);
	tap water, desalinated sea water, at user	ES	0	kg	2.40E-02	1.00E+00	1.22E+00	(2,1,1,1,1,5,BU:1.05);
	plastic tunnel	ES	1	m2a	1.82E-01	1.00E+00	3.06E+00	(3,3,1,1,1,5,BU:3);
	strawberry seedling, for planting, in unheated greenhouse	RER	0	p	8.18E-01	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	polyethylene, HDPE, granulate, at plant	RER	0	kg	2.29E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	polyethylene, LDPE, granulate, at plant	RER	0	kg	7.27E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	steel, low-alloyed, at plant	RER	0	kg	2.79E-05	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	extrusion, plastic pipes	RER	0	kg	1.90E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	extrusion, plastic film	RER	0	kg	7.27E-03	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	blow moulding	RER	0	kg	3.94E-04	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	section bar rolling, steel	RER	0	kg	2.78E-05	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	diesel, burned in agricultural machine	CH	0	kg	7.69E-04	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	electricity, low voltage, at grid	ES	0	kWh	3.25E-02	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	transport, freight, lorry, fleet average	RER	0	tkm	7.80E-03	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
	disposal, steel, 0% water, to municipal incineration	CH	0	kg	2.79E-05	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.26E+02	1.00E+00	1.21E+00	(1,1,1,1,1,5,BU:1.05);
emission air, low population density	Dinitrogen monoxide	-	-	kg	5.00E-05	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	1.85E-04	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Carbon dioxide, fossil	-	-	kg	0.00E+00	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Nitrogen oxides	-	-	kg	1.30E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission air, unspecified	Water, ES	-	-	kg	1.19E+02	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	1.47E-03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, unspecified	Water, ES	-	-	m3	7.72E+03	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, ground-	Phosphorus	-	-	kg	1.27E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Phosphorus	-	-	kg	9.09E-06	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission soil, agricultural	Cadmium	-	-	kg	1.82E-07	1.00E+00	1.32E+00	(4,1,1,1,1,5,BU:1.1);
	Azoxystrobin	-	-	kg	1.09E-06	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Chlorpyrifos methyl	-	-	kg	1.82E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Cyprodinil	-	-	kg	1.27E-06	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Dimethomorph	-	-	kg	7.27E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Etiozazole	-	-	kg	1.82E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Maleic hydrazide	-	-	kg	3.64E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Fosetyl-aluminium	-	-	kg	1.82E-06	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Fludioxonil	-	-	kg	9.09E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Zoxamide	-	-	kg	2.00E-06	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Indoxacarb	-	-	kg	1.82E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Myclobutanil	-	-	kg	3.64E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Penconazole	-	-	kg	1.82E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Pendimethalin	-	-	kg	1.82E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Spinosad	-	-	kg	3.64E-07	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
	Trifloxystrobin	-	-	kg	3.09E-06	1.00E+00	1.30E+00	(2,1,1,1,1,5,BU:1.2);
emission economic-issue, unspecified	Water, embodied in product, ES	-	-	kg	9.00E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 18: Unit process raw data of strawberries, conventional, unheated plastic tunnel/ES

Name	asparagus, conventional
Location	CH
InfrastructureProcess	0
Unit	kg
IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting; mulching and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as the transport to the farm and the direct field emissions.
LocalName	Spargeln, konventionell
Synonyms	
GeneralComment	The inventory refers to the average production of 1 kg of conventional asparagus in Switzerland.
InfrastructureIncluded	1
Category	agricultural production
SubCategory	plant production
LocalCategory	Landwirtschaftliche Produktion
LocalSubCategory	Pflanzenbau
Formula	
StatisticalClassification	
CASNumber	
StartDate	2014
EndDate	2022
DataValidForEntirePeriod	1
OtherPeriodText	
Text	The inventory is modelled for Switzerland
Text	
Percent	
ProductionVolume	
SamplingProcedure	Based on literature and producer interviews
Extrapolations	none
UncertaintyAdjustments	none

Figure 19: Metadata of asparagus, conventional/CH

	Name	Location	Infrastructure Process	Unit	asparagus, conventional	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				CH			
	Infrastructure Process				0			
	Unit				kg			
product	asparagus, conventional	CH	0	kg	1.0	0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.67E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	0.00E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, land	Occupation, annual crop	-	-	m2a	2.81E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, from annual crop, non-irrigated	-	-	m2	2.81E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, to annual crop, non-irrigated	-	-	m2	2.81E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
resource, biotic	Peat	-	-	kg	1.47E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
technosphere	average mineral fertiliser, as N, at regional storehouse	RER	0	kg	3.03E-02	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	average mineral fertiliser, as K2O, at regional storehouse	RER	0	kg	3.54E-02	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	average mineral fertiliser, as P2O5, at regional storehouse	RER	0	kg	1.01E-02	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	magnesium sulphate, at plant	RER	0	kg	7.30E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	pesticide unspecified, at regional storehouse	CH	0	kg	1.63E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	installation for drip irrigation	CH	0	m3	5.34E-03	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	tap water, unspecified natural origin CH, at user	CH	0	kg	5.34E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
	asparagus seedling, open field, at production site	GLO	0	p	8.08E-01	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	polyethylene, LDPE, granulate, at plant	RER	0	kg	1.49E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	extrusion, plastic film	RER	0	kg	1.49E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	electricity, low voltage, production CH, at grid	CH	0	kWh	2.44E-01	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	transport, freight, lorry, fleet average	CH	0	tkm	2.99E-02	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
	disposal, polyethylene, 0.4% water, to municipal incineration	CH	0	kg	1.49E-02	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);
emission air, unspecified	Water, CH	-	-	kg	3.16E+00	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission air, low population density	Dinitrogen monoxide	-	-	kg	5.62E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	1.59E-03	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Nitrogen oxides	-	-	kg	1.18E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	3.23E-02	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	1.97E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Phosphorus	-	-	kg	2.11E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Water, CH	-	-	m3	1.25E-03	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
	Cadmium, ion	-	-	kg	1.07E-08	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	Copper, ion	-	-	kg	8.15E-07	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	Zinc, ion	-	-	kg	2.75E-06	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Lead	-	-	kg	1.80E-08	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Nickel, ion	-	-	kg	1.38E-09	1.00E+00	5.12E+00	(4,1,3,2,1,5,BU:5);
	Chromium VI	-	-	kg	4.78E-06	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
emission soil, agricultural	Cadmium	-	-	kg	5.62E-07	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Nickel	-	-	kg	2.08E-07	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Azoxystrobin	-	-	kg	2.81E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Boscalid	-	-	kg	1.12E-05	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Sodium	-	-	kg	1.57E-03	1.00E+00	1.57E+00	(2,1,2,3,1,5,BU:1.5);
	Cyprodinil	-	-	kg	5.62E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Dimethomorph	-	-	kg	3.51E-07	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Fludioxonil	-	-	kg	2.81E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Fosetyl-aluminium	-	-	kg	1.69E-05	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Metribuzin	-	-	kg	1.83E-05	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Myclobutanil	-	-	kg	3.51E-07	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Pendimethalin	-	-	kg	1.40E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Quinoxifen	-	-	kg	3.51E-07	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Thiacloprid	-	-	kg	7.02E-07	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Trifloxystrobin	-	-	kg	1.05E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
emission economic-issue, unspecified	Water, embodied in product, CH	-	-	kg	9.30E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 20: Unit process raw data of asparagus, conventional/CH

Name	asparagus, conventional
Location	PE
InfrastructureProcess	0
Unit	kg
IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting, mulching and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as the transport to the farm and the direct emissions on the field.
LocalName	Spargeln, konventionell
Synonyms	
GeneralComment	The inventory refers to the average production of 1 kg of conventional asparagus in Peru
InfrastructureIncluded	1
Category	agricultural production
SubCategory	plant production
LocalCategory	Landwirtschaftliche Produktion
LocalSubCategory	Pflanzenbau
Formula	
StatisticalClassification	
CASNumber	
StartDate	2015
EndDate	2022
DataValidForEntirePeriod	1
OtherPeriodText	
Text	The inventory is modelled for Peru.
Text	
Percent	
ProductionVolume	
SamplingProcedure	Based on literature and producer interviews
Extrapolations	none
UncertaintyAdjustments	none

Figure 21: Metadata of asparagus, conventional/PE

	Name	Location	Infrastructure Process	Unit	asparagus, conventional	Uncertainty Type	Standard Deviation 95%	General Comment
	Location				PE			
	Infrastructure Process				0			
	Unit				kg			
product	asparagus, conventional	PE	0	kg	1.0	0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.48E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.39E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, land	Occupation, annual crop	-	-	m2a	1.45E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, from annual crop, non-irrigated	-	-	m2	1.45E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, to annual crop, non-irrigated	-	-	m2	1.45E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
resource, in water	Water, unspecified natural origin, PE	-	-	m3	1.46E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
technosphere	diesel, burned in agricultural machine	CH	0	kg	9.61E-03	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	poultry manure, dried, at regional storehouse	CH	0	kg	5.72E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	vinasse, at fermentation plant	CH	0	kg	7.10E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	ammonium nitrate, as N, at regional storehouse	RER	0	kg	9.25E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	phosphoric acid, fertiliser grade, 70% in H2O, at plant	GLO	0	kg	3.51E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	potassium sulphate, as K2O, at regional storehouse	RER	0	kg	2.17E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	magnesium sulphate, at plant	RER	0	kg	3.99E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	calcium nitrate, as N, at regional storehouse	RER	0	kg	1.20E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	boric acid, anhydrous, powder, at plant	RER	0	kg	8.55E-04	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	urea ammonium nitrate, as N, at regional storehouse	RER	0	kg	1.30E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	potassium nitrate, as N, at regional storehouse	RER	0	kg	3.22E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	pesticide unspecified, at regional storehouse	CH	0	kg	1.44E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	installation for drip irrigation	CH	0	m3	1.46E+00	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	asparagus seedling, open field, at production site	GLO	0	p	8.33E-01	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	electricity, low voltage, production PE, at grid	PE	0	kWh	5.31E-01	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	transport, freight, lorry, fleet average	RER	0	tkm	2.91E-02	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
emission air, unspecified	Water, PE	-	-	kg	6.79E+02	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission air, low population density	Dinitrogen monoxide	-	-	kg	4.80E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	1.78E-03	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Carbon dioxide, fossil	-	-	kg	8.78E-03	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Nitrogen oxides	-	-	kg	1.01E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	2.36E-02	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	1.01E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, unspecified	Water, PE	-	-	m3	7.85E-01	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Cadmium, ion	-	-	kg	5.65E-10	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Copper, ion	-	-	kg	3.91E-07	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Zinc, ion	-	-	kg	1.74E-06	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Lead	-	-	kg	1.35E-08	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Nickel, ion	-	-	kg	1.20E-09	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Chromium, ion	-	-	kg	1.45E-06	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Mercury	-	-	kg	2.32E-11	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Phosphorus	-	-	kg	1.07E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission soil, agricultural	Nickel	-	-	kg	4.49E-07	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Mercury	-	-	kg	9.57E-10	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Chlorpyrifos	-	-	kg	1.19E-03	1.00E+00	1.30E+00	(2,1,2,1,1,5,BU:1.2);
	Methomyl	-	-	kg	2.17E-04	1.00E+00	1.30E+00	(2,1,2,1,1,5,BU:1.2);
	Sulfur	-	-	kg	7.25E-06	1.00E+00	1.57E+00	(2,1,2,2,1,5,BU:1.5);
	Tebuconazole	-	-	kg	2.54E-05	1.00E+00	1.30E+00	(2,1,2,1,1,5,BU:1.2);
emission economic-issue, unspecified	Water, embodied in product, PE	-	-	kg	9.30E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

Figure 22: Unit process raw data of asparagus, conventional/PE

Name	asparagus, conventional
Location	MX
InfrastructureProcess	0
Unit	kg
IncludedProcesses	The inventory includes irrigating, fertilisation, pest and pathogen control, harvesting; mulching and the machines and shed or surface used to park them; all inputs as young plants (seedlings), fertilizers (mineral and organic), growing media, active substances, water for irrigation, fuels end energy as well as the transport to the farm and the direct field emissions.
Amount	1
LocalName	Spargeln, konventionell
Synonyms	
GeneralComment	The inventory refers to the average production of 1 kg of conventional asparagus in Mexico.
InfrastructureIncluded	1
Category	agricultural production
SubCategory	plant production
LocalCategory	Landwirtschaftliche Produktion
LocalSubCategory	Pflanzenbau
Formula	
StatisticalClassification	
CASNumber	
StartDate	2015
EndDate	2022
DataValidForEntirePeriod	1
OtherPeriodText	
Text	The inventory is modelled for Mexico.
Text	
Percent	
ProductionVolume	
SamplingProcedure	Based on literature and producer interviews
Extrapolations	none
UncertaintyAdjustments	none

Figure 23: Metadata of asparagus, conventional/MX

	Name	Location	Infrastructure Process	Unit	asparagus, conventional	Uncertainty Type	Standard Deviation 95%	General Comment
	Location Infrastructure Process Unit				MX 0 kg 1.0			
product	asparagus, conventional	MX	0	kg		0		
resource, biotic	Energy, gross calorific value, in biomass	-	-	MJ	1.67E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, in air	Carbon dioxide, in air	-	-	kg	1.39E-01	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, land	Occupation, annual crop	-	-	m2a	1.61E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
resource, in water	Water, unspecified natural origin, MX	-	-	m3	1.63E+00	1.00E+00	1.32E+00	(4,1,3,2,1,5,BU:1.05);
resource, land	Transformation, from annual crop, non-irrigated	-	-	m2	1.61E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
	Transformation, to annual crop, non-irrigated	-	-	m2	1.61E+00	1.00E+00	1.34E+00	(4,1,3,2,1,5,BU:1.1);
technosphere	diesel, burned in agricultural machine	CH	0	kg	1.03E-02	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	poultry manure, dried, at regional storehouse	CH	0	kg	1.10E-02	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	vinasse, at fermentation plant	CH	0	kg	5.50E-03	1.00E+00	1.22E+00	(2,1,2,1,1,5,BU:1.05);
	ammonium nitrate, as N, at regional storehouse	RER	0	kg	81.5/6200	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	phosphoric acid, fertiliser grade, 70% in H2O, at plant	GLO	0	kg	3.90E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	potassium sulphate, as K2O, at regional storehouse	RER	0	kg	1.89E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	magnesium sulphate, at plant	RER	0	kg	4.45E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	calcium nitrate, as N, at regional storehouse	RER	0	kg	1.48E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	boric acid, anhydrous, powder, at plant	RER	0	kg	1.38E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	urea ammonium nitrate, as N, at regional storehouse	RER	0	kg	1.01E-02	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	potassium nitrate, as N, at regional storehouse	RER	0	kg	5.92E-03	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	pesticide unspecified, at regional storehouse	CH	0	kg	9.92E-04	1.00E+00	1.22E+00	(2,1,2,3,1,5,BU:1.05);
	installation for drip irrigation	CH	0	m3	1.63E+00	1.00E+00	3.10E+00	(4,1,3,2,1,5,BU:3);
	asparagus seedling, open field, at production site	GLO	0	p	9.27E-01	1.00E+00	1.69E+00	(4,4,4,2,4,5,BU:1.05);
	electricity, low voltage, production MX, at grid	MX	0	kWh	5.51E-01	1.00E+00	1.22E+00	(2,1,1,3,1,5,BU:1.05);
	transport, freight, lorry, fleet average	RER	0	tkm	7.17E-02	1.00E+00	2.05E+00	(2,1,1,3,1,5,BU:2);
emission air, unspecified	Water, MX	-	-	kg	7.56E+02	1.00E+00	1.63E+00	(4,1,3,2,1,5,BU:1.5);
emission air, low population density	Dinitrogen monoxide	-	-	kg	5.63E-04	1.00E+00	1.57E+00	(2,2,2,1,1,5,BU:1.5);
	Ammonia	-	-	kg	1.79E-03	1.00E+00	1.30E+00	(2,2,2,1,1,5,BU:1.2);
	Carbon dioxide, fossil	-	-	kg	6.82E-03	1.00E+00	1.22E+00	(2,2,2,1,1,5,BU:1.05);
	Nitrogen oxides	-	-	kg	1.18E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, ground-	Nitrate	-	-	kg	2.82E-02	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
	Phosphorus	-	-	kg	1.13E-05	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, unspecified	Water, MX	-	-	m3	8.72E-01	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission water, river	Cadmium, ion	-	-	kg	6.13E-09	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Copper, ion	-	-	kg	4.35E-07	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Zinc, ion	-	-	kg	2.26E-06	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Lead	-	-	kg	1.52E-08	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Nickel, ion	-	-	kg	1.42E-09	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Chromium, ion	-	-	kg	2.58E-06	1.00E+00	3.09E+00	(4,1,1,1,1,5,BU:3);
	Mercury	-	-	kg	3.06E-11	1.00E+00	5.10E+00	(4,1,1,1,1,5,BU:5);
	Phosphorus	-	-	kg	1.21E-04	1.00E+00	1.62E+00	(4,1,1,1,1,5,BU:1.5);
emission soil, agricultural	Cadmium	-	-	kg	3.06E-07	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Nickel	-	-	kg	7.58E-07	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Mercury	-	-	kg	1.34E-09	1.00E+00	1.26E+00	(2,1,3,1,1,5,BU:1.1);
	Boscalid	-	-	kg	8.06E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Flonicamid	-	-	kg	1.61E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Fludioxonil	-	-	kg	4.03E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Imidacloprid	-	-	kg	4.03E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Metalaxyl-M	-	-	kg	4.03E-06	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
	Glyphosate	-	-	kg	1.33E-04	1.00E+00	1.30E+00	(2,1,2,3,1,5,BU:1.2);
emission economic-issue, unspecified	Water, embodied in product, MX	-	-	kg	9.30E-01	1.00E+00	1.30E+00	(4,1,1,1,1,5,BU:1.05);

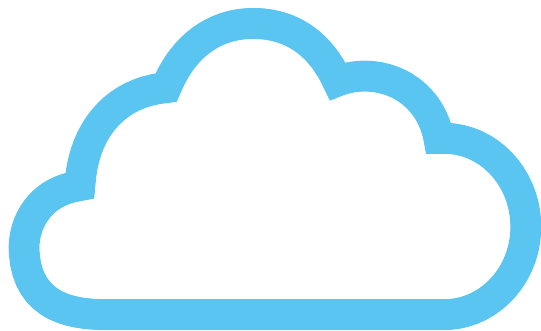
Figure 24: Unit process raw data of asparagus, conventional/MX

A5 Review report

External review

LCA of strawberry and asparagus production

Bundesamt für Umwelt BAFU



Content: External Review of the “Life cycle assessment of strawberry and asparagus production”, carried out by Carbotech AG and commissioned by the Bundesamt für Umwelt BAFU

Date: 26.08.2022

Reviewer: Andy Eigenmann
Project Manager Consulting and Solutions
+41 (0)44 500 43 50
andy.eigenmann@myclimate.org

Client: Bundesamt für Umwelt BAFU
Mr. Pierryves Padey
Papiermühlestrasse 172
3003 Bern

Content

1	Background.....	2
2	Reviewed content.....	2
3	Guidelines.....	2
4	Review.....	3
4.1	Completeness of the documentation	3
4.2	Consistency with the quality guidelines	3
4.3	Plausibility check of the life cycle inventory data	4
4.4	Completeness of inputs and outputs	4
4.5	Mathematical correctness of calculations	4
5	Conclusion from the reviewer	5

1 Background

The Bundesamt für Umwelt BAFU has commissioned Carbotech AG to generate the inventory data and life cycle assessment calculations of strawberries and asparagus. So far, the existing inventories of tomatoes and beans were often used as examples from the agricultural and food sector. Now, these are to be expanded by other products in the field of vegetables and fruits, to further raise the awareness of customer decisions and their impact on the environment.

2 Reviewed content

The validation is based on the following data, provided by Thomas Kägi of Carbotech AG:

- Final version of the report “Life cycle assessment of strawberry and asparagus production” from the 14th August 2022.
- All main and auxiliary inventories newly created and used for the study.

3 Guidelines

The validation of the study at hand is guided by the reference provided in ecoinvent report No.1 “Overview and Methodology”¹, especially the review principles listed in chapter 9.4:

Completeness of the documentation. All investigated datasets should be described in the report, and all necessary meta information and flow data should be available for each dataset.

Consistency with the quality guidelines. It is checked whether the unit processes have been modelled according to the ecoinvent quality guidelines. The quality guidelines cover for example the estimation of transport or the calculation of energy demands in the inventory

Plausibility check of the life cycle inventory data. Selected input and output flows are controlled for plausibility.

Completeness of inputs and outputs. The completeness of flows is based on the environmental and technical knowledge of the reviewing person. Reviewers are not necessarily technical experts of the processes reviewed. If necessary they were supported by the person responsible for the report.

Mathematical correctness of calculations. Selected inputs and outputs are controlled in view of mathematical correctness, e.g. the transport service inputs, the waste heat or CO₂ emissions.

Furthermore, it was also verified that the study complies with the international standards for life cycle assessments (ISO 14040 and 14044).

¹ Frischknecht, R., & Jungbluth, N. (2007). Overview and Methodology. Data v2.0. ecoinvent report no. 1. Dübendorf: Swiss Centre for Life Cycle Inventories.

4 Review

4.1 Completeness of the documentation

The present study report is very detailed and well documented. The background of the study, as well as the goal and scope are clearly described. A summary of all considered variants of strawberries and asparagus is already provided at the beginning, which simplifies the overview.

The functional unit used, and the system boundaries of the study are described in an understandable way.

All inventories newly created for the study are listed and described. The description itself is very detailed and lists all the data used and their sources in detail. In addition, other data sources that were used for plausibility checks are listed and the differences to the data used in the study are explained. On the one hand, this provides a very good overview, and on the other hand, it allows the background data of the study to be put into context. Furthermore, the assumptions made by Carbotech to fill data gaps in the inventories are also well documented.

The results of the study are all listed and presented in graphical form, which makes it easy to compare the different product variants. The discussion is going into detail regarding the results, analyzing the different life phases and putting them into context.

Summary

In general, the documentation has a very high level of detail. It is comprehensible, transparent and covers all areas of the study. It therefore meets the criterion of completeness.

4.2 Consistency with the quality guidelines

The quality guidelines include the validation of underlying scope and modelling principles, the documentation of elementary flows, dealing with multi-output and allocation rules, and uncertainty considerations.

The underlying scope was appropriately chosen for this study. The geographical focus lies on the one hand on a cultivation in Switzerland, and on the other hand on various other traditional growing countries, which allows a good comparison. In the case of the basic data used, attention was paid to using the most up-to-date sources available. When modelling data gaps and uncertainties, comparative values were used if available, or an approach was taken based on real situations or well founded assumptions.

All elementary flows that were newly modelled are noted in the study report. The division into the different categories and subcategories was also carried out correctly. If changes were made to elementary flows in existing inventories, this is also documented in the report and is therefore traceable.

Multi-output processes are not present in this study and no allocations had to be made.

As in most cases only one source was available for the data for newly compiled inventories, which contained only single or average values, no detailed uncertainty calculation could be carried out. Therefore, an uncertainty assessment was carried out using the Pedigree matrix, which is in accordance with the quality guideline.

Summary

The study was conducted in accordance with the four areas of the ecoinvent quality guideline, and thus meets these criteria.

4.3 Plausibility check of the life cycle inventory data

On the one hand, the plausibility of the data was checked in a joint meeting with the authors. The data used were checked, while additional detailed information was provided by the authors. A first verification of the inventories and results could already be done.

On the other hand, an in-depth review of the data sets was carried out after this meeting, in combination with the report. Thereby, the inputs and outputs were checked on a sample basis and compared with other sources.

Summary

No inconsistencies or serious deviations were identified during the review. The available data are plausible and well founded. The criterion of plausibility is therefore met.

4.4 Completeness of inputs and outputs

The inputs and outputs of the present study have a high degree of completeness. Especially the highly relevant areas, such as pesticide use in non-European countries, have been included in detail. Inputs and outputs in the cultivation phase are generally extensively covered, which is very relevant due to the importance of this phase.

The flows in the other life phases are also well modelled. The use of materials in infrastructure and packaging, their disposal and the various transport processes were fully covered.

Summary

All relevant inputs and outputs were recorded in full and included in this study. The validation criterion of completeness of inputs and outputs is therefore fulfilled.

4.5 Mathematical correctness of calculations

During the plausibility check on site together with the authors, an initial review of the calculations was also carried out. This was followed by a second, more detailed examination of the underlying values and formulas. During this, all calculations could be verified, with no errors present and the modelling set up being correct.

Summary

The mathematical basis of the present study is correct, and the criterion is therefore fulfilled.

5 Conclusion from the reviewer

The present study fully complies with all criteria of the ecoinvent guideline, as well as the ISO standards 14040/44. It has a high level of detail, especially in the very relevant phase of cultivation. Moreover, it was carefully carried out and well documented.

During the review, the reviewer was provided with all necessary data and documents, and all questions and uncertainties were immediately considered and discussed.

The reviewer therefore expects that the inventories created can be included in the database.

Zürich, 26.08.2022



Andy Eigenmann